Solutions to Algebraic Curves - An Introduction to Algebraic Geometry - William Fulton

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March 1, 2026

Preface

The goal of this document is to share my personal solutions to the exercises in the book Algebraic Curves - An Introduction to Algebraic Geometry by William Fulton during my reading.
As a disclaimer, the solutions are not unique and there will probably be better or more optimized solutions than mine. Feel free to correct me or ask me anything about the content of this document at the following address :
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Chapter 1

Affine Algebraic Sets

1.1 Algebraic Preliminaries

1.1. Let R be a domain. (a) If F, G are forms of degree r, s respectively in $R[X_1, ..., X_n]$, show that FG is a form of degree $r + s$. (b) Show that any factor of a form in $R[X_1, ..., X_n]$ is also a form.

Solution

(a) The form F is a sum of monomials of degree r and G is a sum of monomials of degree s. Hence, using the definition of the product of two polynomials, the polynomial FG is a sum of products ab where a is a monomial of degree r and b is a monomial of degree s. Hence, ab is a monomial of degree $r + s$ so FG is a sum of monomials of degree $r + s$. Therefore, equivalently, FG is a form of degree $r + s$.

(b) Let f be a divisor of a form F of degree N, then $F = fg$ for some polynomial g. Suppose that f is not a form, then we can write $f = f_a + f_{a+1} + \cdots + f_b$ where f_i is a form of degree i, $a < b$, and $f_a, f_b \neq 0$. Similarly, we can write $g = g_c + g_{c+1} + \cdots + g_d$ where g_i is a form of degree i, $c \leq d$, and $g_c, g_d \neq 0$. It follows that

$$F = fg = f_a g_c + [f_a g_{c+1} + f_{a+1} g_c] + \cdots + f_b g_d$$

is the decomposition of F into a sum of forms. However, $f_a g_c \neq 0$ is a form of degree $a + c$ and $f_b g_d \neq 0$ is a form of degree $b + d > a + c$. But this impossible because F is already a form so it cannot be decomposed into nonzero forms of different degree.

1.2. Let R be a UFD, K the quotient field of R. Show that every element z of K may be written $z = a/b$, where $a, b \in R$ have no common factors; this representative is unique up to units of R.

Solution Let $z \in K$ and write it as a/b with $a, b \in R$. Since R is a UFD, then we can define $d = \gcd(a, b)$, $a = a'd$ and $b = b'd$. Since d is the greatest common divisor of a and b, then a' and b' have no common divisor. Hence, $z = a/b = a'/b'$ where the numerator and the denominator have no common divisor. Next, we need

to show that this representation of z is unique up to multiplication by units. If c/d is another representation of z as a quotient of two elements in R with $\gcd(c,d)=1$, then $a'/b' = c/d$. From this equation, we get that $a'd = b'c$. Hence, a' divides $b'c$, but $\gcd(a,b)=1$ so a' divides c. Similarly, c divides $a'd$ but $\gcd(c,d)=1$ so c divides a'. Thus, $c' = ua'$ where u is a unit. If we plug this into the equation $a'd = b'c$, we get that $a'd = b'ua'$, and hence, $d = ub'$. Therefore, c and d are equal to a and b, up to multiplication by a unit.

1.3. Let R be a PID, let P be a nonzero, proper, prime ideal in R. (a) Show that P is generated by an irreducible element. (b) Show that P is maximal.

Solution

(a) Since R is a PID, then $P = (p)$ where $p \in R$. Since P is nonzero, and proper, then p is not zero, nor a unit. Suppose that p is reducible, then there exist two non-units and nonzero elements $a, b \in R$ such that $p = ab$. Since P is prime, then $ab = p \in (p)$ implies that $a \in (p)$ or $b \in (p)$. If $a \in (p)$, p divides a. Since a divides p, then $ab = p = au$ where u is a unit, so $b = u$, a contradiction. Therefore, p is irreducible.

(b) Let $I = (f)$ be an ideal containing P, then $p \in (f)$ which implies that f divides p. But the only divisors of p are units and unit multiples of p. In the first case, $I = R$, in the second case, $I = P$. Therefore, P is maximal.

1.4. Let k be an infinite field, $F \in k[X_1, ..., X_n]$. Suppose $F(a_1, \ldots, a_n) = 0$ for all $a_1, \ldots, a_n \in k$. Show that $F = 0$. (*Hint:* Write $F = \sum F_i X_n^i$, $F_i \in k[X_1, ..., X_{n-1}]$. Use induction on n, and the fact that $F(a_1, ..., a_{n-1}, X_n)$ has only a finite number of roots if any $F_i(a_1, ..., a_{n-1}) \neq 0$.)

Solution Let's prove this by induction on n. When F is a polynomial in $k[X]$ such that $F(a) = 0$ for all $a \in k$, then $F = 0$ because F has at most finitely many roots when it is nonzero. Next, suppose that $G \in k[X_1, ..., X_{n-1}]$ with $G(a_1, ..., a_{n-1}) = 0$ for all $(a_1, ..., a_{n-1}) \in k^{n-1}$ implies that $G = 0$, and consider a polynomial $F \in k[X_1, ..., X_n]$ with $F(a_1, ..., a_n) = 0$ for all $(a_1, ..., a_n) \in k^n$. If we view F as an elements of $k[X_1, ..., X_{n-1}][X_n]$, then we can write $F = \sum_i F_i X_n^i$ where $F_i \in k[X_1, ..., X_{n-1}]$. Fix $(a_1, ..., a_{n-1}) \in k^{n-1}$ and consider the polynomial $f(X) = F(a_1, ..., a_{n-1}, X) = \sum_i F_i(a_1, ..., a_{n-1}) X^i$, then $f(a_n) = 0$ for all a. By the case $n = 1$, we have that $f = 0$, and hence, that $\sum_i F_i(a_1, ..., a_{n-1}) X^i = 0$. By properties of polynomials, it follows that $F_i(a_1, ..., a_{n-1}) = 0$ for all i. Since it holds for all $(a_1, ..., a_{n-1}) \in k^{n-1}$, then by induction, $F_i = 0$ for all i. Hence, $F = \sum_i F_i X_n^i = 0$. Therefore, by induction, it holds for all n.

1.5. Let k be any field. Show that there are an infinite number of irreducible monic polynomials in $k[X]$. (*Hint:* Suppose $F_1, ..., F_n$ were all of them, and factor $F_1 \cdots F_n + 1$ into irreducible factors.)

Solution Suppose that there are finitely many irreducible monic polynomials $F_1, ..., F_n$. Define the polynomial $F = F_1 \cdots F_n + 1$, then this polynomial must

be divisible by F_i for some i. However, F_i certainly divides $F_1 \cdots F_n$ so F_i must divide $F - F_1 \cdots F_n = 1$. But this is a contradiction since the only monic polynomial dividing 1 is 1, which is not irreducible since it is a unit. Therefore, by contradiction, there are infinitely many irreducible polynomials.

1.6. Show that any algebraically closed field is infinite. (*Hint:* The irreducible monic polynomials are $X - a$, $a \in k$.)

Solution Clearly, every polynomial of the form $X - a$ with $a \in k$ is irreducible. Moreover, since k is algebraically closed, then every polynomial can be factored into a product of linear factors so the irreducible polynomials are precisely the polyomials of the form $X - a$ with $a \in k$. It is easy to see then that there are as many irreducible polynomials as elements in k. In the previous Problem, it is shown that there are always infinitely many irreducible polynomials. Therefore, k is infinite.

1.7. Let k be a field, $F \in k[X_1, ..., X_n]$, $a_1, ..., a_n \in k$. (a) Show that

$$F = \sum \lambda_{(i)}(X_1 - a_1)^{i_1} \cdots (X_n - a_n)^{i_n}, \quad \lambda_{(i)} \in k.$$

(b) If $F(a_1, ..., a_n) = 0$, show that $F = \sum_{i=1}^{n}(X_i - a_i)G_i$ for some (not unique) G_i in $k[X_1, ..., X_n]$.

Solution

(a) Notice that $F = F(X_1 + a_1 - a_1, ..., X_n + a_n - a_n)$. Hence, if we let $G = F(X_1 + a_1, ..., X_n + a_n) \in k[X_1, ..., X_n]$, then we can write

$$G = \sum \lambda_{(i)} X_1^{i_1} \cdots X_n^{i_n},$$

then

$$F = G(X_1 - a_1, ..., X_n - a_n) = \sum \lambda_{(i)}(X_1 - a_1)^{i_1} \cdots (X_n - a_n)^{i_n}$$

for some $\lambda_{(i)} \in k$.

(b) If $F(a_1, ..., a_n) = 0$, then $G(0, ..., 0) = 0$, so G has no constant term. Let's prove that G can be written as $\sum X_i H_i$ where $H_i \in k[X_1, ..., X_n]$. We can write G as $X_1 H_1 + H_1'$ where none of the monomials in H_1' contain a factor of X_1. Next, write H_1' as $X_2 H_2 + H_2'$ where none of the monomials in H_2' contain a factor of X_1 or X_2. If we repeat this process, we get that $G = \sum X_i H_i$. It follows that

$$F = \sum (X_i - a_i) H_i (X_1 - a_1, ..., X_n - a_n).$$

1.2 Affine Space and Algebraic Sets

1.8. Show that the algebraic subsets of $\mathbb{A}^1(k)$ are just the finite subsets, together with $\mathbb{A}^1(k)$ itself.

Solution Let S be an arbitrary set polynomials in $k[x]$. If S is empty or $S = \{0\}$, then $V(S) = \mathbb{A}^1(k)$. If S contains a nonzero polynomial F, then $V(S) \subset V(F)$.

Since F is a one-variable polynomial, then it has finitely many roots, and so $V(F)$ is finite. It follows that $V(S)$ is a finite subset of $\mathbb{A}^1(k)$. Conversely, we know that any finite subsets are algebraic sets. Therefore, the algebraic subsets of $\mathbb{A}^1(k)$ are precisely the finite subsets and $\mathbb{A}^1(k)$ itself.

1.9. If k is a finite field, show that every subset of $\mathbb{A}^1(k)$ is algebraic.

Solution Let U be a subset of $\mathbb{A}^n(k)$, then U us finite and so it must be algebraic by property n°5 of the section.

1.10. Give an example of a countable collection of algebraic sets whose union is not algebraic.

Solution If we let $k = \mathbb{R}$, and $V_i = \{i\}$ for all $i \in \mathbb{Z}$, then all the V_i's are algebraic subsets of $A^1(k)$ but their union is not since it would contradict Problem 1.8.

1.11. Show that the following are algebraic sets:

(a) $\{(t, t^2, t^3) \in \mathbb{A}^3(k) | t \in k\}$;

(b) $\{(\cos(t), \sin(t)) \in \mathbb{A}^2(\mathbb{R}) | t \in \mathbb{R}\}$;

(c) the set of points in $\mathbb{A}^2(\mathbb{R})$ whose polar coordinates (r, θ) satisfy the equation $r = \sin(\theta)$.

Solution

(a) Let $U = \{(t, t^2, t^3) \in \mathbb{A}^3(k) | t \in k\}$ and $V = V(x^3 - z, x^2 - y)$. Let $(x, y, z) \in U$, then there is a $t \in k$ such that $x = t$, $y = t^2$, and $z = t^3$. It follows that $x^3 - z = 0$ and $x^2 - y = 0$. Hence, $(x, y, z) \in V$. Conversely, if $(x, y, z) \in V$, and if we let $t = x$, then $y = x^2 = t^2$ and $z = x^3 = t^3$. Hence, $(x, y, z) \in U$. Therefore, $U = V$ and so U is algebraic.

(b) Since the set $U = \{(\cos(t), \sin(t)) \in \mathbb{A}^2(\mathbb{R}) | t \in \mathbb{R}\}$ is simply the unit circle in $\mathbb{R}^2$, then we know that we can rewrite it as $\{(x, y) \in \mathbb{A}^2(\mathbb{R}) | x^2 + y^2 = 1\}$, or simply as $V(x^2 + y^2 - 1)$. Thus, it is an algebraic set.

(c) Using the fact that $r = \sqrt{x^2 + y^2}$ and $\sin(\theta) = y/\sqrt{x^2 + y^2}$, we get that the set $\{(x, y) \in \mathbb{A}^2(\mathbb{R}) | r = \sin(\theta)\}$ is equal to

$$\{(x, y) \in \mathbb{A}^2(\mathbb{R}) | \sqrt{x^2 + y^2} = y/\sqrt{x^2 + y^2}\}.$$

Equivalently, we can rewrite it as $V(x^2 + y^2 - y)$. Hence, it is an algebraic set.

1.12. Suppose C is an affine plane curve, and L is a line in $\mathbb{A}^2(k)$, $L \not\subset C$. Suppose $C = V(F)$, $F \in k[X, Y]$ a polynomial of degree n. Show that $L \cap C$ is a finite set of no more than n points. (*Hint:* Suppose $L = V(Y - (aX + b))$, and consider $F(X, aX + b) \in k[X]$.)

Solution Let $F \in k[X, Y]$ be a polynomial of degree n and suppose that $C = V(F)$. Since L is a line, then we can write $L = V(G)$ with $G = aX + bY + c$ for some

$a, b, c \in k$ where a and b cannot be both zero. Suppose without loss of generality that b is non-zero, then $(X, Y) \in L$ if and only if $Y = -\frac{a}{b}X - \frac{c}{b}$. Consider now the polynomial $F(X, -\frac{a}{b}X - \frac{c}{b}) \in k[X]$, then this is a polynomial of degree n. Let X be one of its roots, then $(X, -\frac{a}{b}X - \frac{c}{b}) \in L \cap C$. Conversely, if $(X, Y) \in L \cap C$, then $Y = -\frac{a}{b}X - \frac{c}{b}$ which implies that $F(X, Y) = 0$ and so X is a root of $F(X, -\frac{a}{b}X - \frac{c}{b})$. Therefore, the roots of X are in bijection with the elements in $L \cap C$. Since $F(X, -\frac{a}{b}X - \frac{c}{b})$ is a polynomial of degree n, then it has at most n roots, and hence, $L \cap C$ is a finite set of no more than n points.

1.13. Show that each of the following sets is not algebraic:

(a) $\{(x, y) \in \mathbb{A}^2(\mathbb{R}) \mid y = \sin(x)\}$.

(b) $\{(z, w) \in \mathbb{A}^2(\mathbb{C}) \mid |z|^2 + |w|^2 = 1\}$, where $|x + iy|^2 = x^2 + y^2$ for $x, y \in \mathbb{R}$.

(c) $\{(\cos(t), \sin(t), t) \in \mathbb{A}^3(\mathbb{R}) | t \in \mathbb{R}\}$;

Solution

(a) Suppose that $\{y = \sin(x)\} = V(S)$ for some $S \subset k[X, Y]$. Since $\{y = \sin(x)\} \neq \mathbb{A}^2(\mathbb{R})$, then S must contain a nonzero polynomial $F \in k[X, Y]$, of degree n. Since $F \in S$, then $F(x, \sin(x)) = 0$ for all $x \in \mathbb{R}$. Consider now the polynomial $F(0, X) \in k[X]$, then this polynomial must have at most n roots. However, there are infinitely many $x \in \mathbb{R}$ such that $\sin(x) = 0$ since F has infinitely many roots. Therefore, by contradiction, the set $\{y = \sin(x)\}$ is not algebraic.

(b) Suppose that $\{|z|^2 + |w|^2 = 1\} = V(S)$, since $\{|z|^2 + |w|^2 = 1\} \neq \mathbb{A}^2(\mathbb{C})$, then there must be a nonzero polynomial $F \in k[X, Y]$ such that $F(z, w) = 0$ for all $z, w \in \mathbb{C}$ such that $|z|^2 + |w|^2 = 1$. Let $C = V(F)$, and let $L = V(X)$ be a line. We know that $L \not\subset C$ since $(0, 2) \in L$ but $(0, 2) \notin C$. Hence, by Problem 1.12, $L \cap C$ is finite. However, for every complex number w with $|w| = 1$, we have that $(0, w) \in L \cap C$. Since there are infinitely many such complex numbers w, then $L \cap C$ is infinite, a contradiction. Therefore, $\{|z|^2 + |w|^2 = 1\}$ is not an algebraic set.

(c) Suppose that $\{(\cos(t), \sin(t), t)\}_{t \in \mathbb{R}} = V(S)$ for some $S \subset k[X, Y, Z]$. Since $\{(\cos(t), \sin(t), t)\}_{t \in \mathbb{R}} \neq \mathbb{A}^3(\mathbb{R})$, then S must contain a nonzero polynomial $F \in k[X, Y, Z]$, of degree n. Since $F \in S$, then $F(\cos(t), \sin(t), t) = 0$ for all $t \in \mathbb{R}$. Consider now the polynomial $F(1, 0, X) \in k[X]$, then this polynomial must have at most n roots. However, there are infinitely many $t \in \mathbb{R}$ such that $\cos(x) = 1$, $\sin(x) = 0$, and hence, $F(1, 0, x) = F(\cos(x), \sin(x), x) = 0$, so F has infinitely many roots. Therefore, by contradiction, the set is not algebraic.

1.14. Let F be a nonconstant polynomial in $k[X_1, \ldots, X_n]$, k is algebraically closed. Show that $\mathbb{A}^n(k) \setminus V(F)$ is infinite if $n \geq 1$, and $\mathbb{A}^n(k) \setminus V(F)$ is infinite if $n \geq 2$. Conclude that the complement of any proper algebraic set is infinite. (*Hint:* See Problem 1.4.)

Solution First, notice that k must be infinite since it is algebraically closed. Hence, the case $n = 1$ follows from Problem 1.8. For the case $n = 2$, suppose that there are

finitely many (x, y) such that $F(x, y) \neq 0$. Let (x_0, y_0) be such a pair, then we get that there finitely many y such that $F(x_0, y) = 0$. But since $F(x_0, y) \in k[y]$, then we get a contradiction with the case $n = 1$. Therefore, $\mathbb{A}^n(k) \setminus V(F)$ is infinite for the case $n = 2$.

Now, let $n \geq 1$, $F \in k[X_1, ..., X_n]$, and suppose that $\mathbb{A}^n(k) \setminus V(F)$ is finite. Let $(a_1, ..., a_n) \in \mathbb{A}^n(k)$ be such that $F(a_1, ..., a_n) \neq 0$, then the polynomial $F(a_1, ..., a_{n-1}, X) \in k[X]$ is such that there are finitely many $x \in k$ such that it is nonzero. But this is in contradiction with the case $n = 1$. Thus, by contradiction, the set $\mathbb{A}^n(k) \setminus V(F)$ is infinite.

1.15. Let $V \subset \mathbb{A}^n(k)$, $W \subset \mathbb{A}^m(k)$ be algebraic sets. Show that

$$V \times W = \{(a_1, ..., a_n, b_1, ..., b_m) \mid (a_1, ..., a_n) \in V, (b_1, ..., b_m) \in W\}$$

is an algebraic set in $\mathbb{A}^{n+m}(k)$. It is called the *product* of V and W.

Solution For all $F \in V$ and $G \in W$, define $F^\times, G^\times \in k[X_1, ..., X_{n+m}]$ by

$$F^\times(X_1, ..., X_{n+m}) = F(X_1, ..., X_n) \text{ and } G^\times(X_1, ..., X_{n+m}) = G(X_{n+1}, ..., X_{n+m}).$$

If we consider the set $V(\{F^\times\}_{F \in V})$, we get that it is equal to the elements in $\mathbb{A}^{n+m}(k)$ where the first n entries are in V. Similarly, $V(\{G^\times\}_{G \in W})$ is equal to the elements in $\mathbb{A}^{n+m}(k)$ where the last m entries are in V. It follows that $V(\{F^\times\}_{F \in V} \cup \{G^\times\}_{G \in W}) = V \times W$. Therefore, $V \times W$ is an algebraic set.

1.3 The Ideal of a Set of Points

1.16. Let V, W be algebraic sets in $\mathbb{A}^n(k)$. Show that $V = W$ if and only if $I(V) = I(W)$.

Solution If $V = W$, then $I(V) = I(W)$ trivially. Now, if $I(V) = I(W)$, then $V(I(V)) = V(I(W))$. Since V and W are algebraic sets, then $V(I(V)) = V$ and $V(I(W)) = W$. Hence, $V = W$.

1.17. (a) Let V be an algebraic set in $\mathbb{A}^n(k)$, $P \in \mathbb{A}^n(k)$ a point not in V. Show that there is a polynomial $F \in k[X_1, ..., X_n]$ such that $F(Q) = 0$ for all $Q \in V$, but $F(P) = 1$. (*Hint:* $I(V) \neq I(V \cup \{P\})$.) (b) Let $P_1, ..., P_r$ be distinct points in $\mathbb{A}^n(k)$, not in an algebraic set V. Show that there are polynomials $F_1, ..., F_r \in I(V)$ such that $F_i(P_j) = 0$ if $i \neq j$, and $F_i(P_i) = 1$. (*Hint:* Apply (a) to the union of V and all but one point.) (c) With $P_1, ..., P_r$ and V as in (b), and $a_{ij} \in k$ for $1 \leq i, j \leq r$, show that there are $G_i \in I(V)$ with $G_i(P_j) = a_{ij}$ for all i and j. (*Hint:* Consider $\sum_j a_{ij} F_j$.)

Solution

(a) Since $P \notin V$, then $V \neq V \cup \{P\}$ and so $I(V) \neq I(V \cup \{P\})$ by Problem 1.16. Since $V \subset V \cup \{P\}$, then $I(V \cup \{P\}) \subset I(V)$ and so it must be that $I(V) \not\subset I(V \cup \{P\})$. This implies that there is a $F_0 \in I(V) \subset k[X_1, ..., X_n]$ such that $F_0 \notin I(V \cup \{P\})$. In other words, there is polynomial $F_0 \in k[X_1, ..., X_n]$ such that $F_0(Q) = 0$ for all $Q \in V$ and $F_0(P) \neq 0$. Define $F(x) = F_0(P)^{-1} F_0(x)$ for all $x \in \mathbb{A}^n(k)$, then $F(Q) = 0$ for all $Q \in V$ and $F(P) = 1$.

(b) Let $i \in [\![1, r]\!]$, since $V \cup \{P_j\}_{j \neq i}$ is an algebraic set that doesn't contain P_i, then by part (a) there is a function $F_i \in I(V)$ such that $F_i(P_j) = 0$ whenever $j \neq i$ and $F_i(P_i) = 1$.

(c) Since the set up is the same as in (b), then we already showed that, for all $i \in [\![1, r]\!]$, there exists a polynomial $F_i \in I(V)$ such that $F_i(P_j) = 0$ for all $j[\![1, r]\!] \setminus \{i\}$ and $F_i(P_i) = 1$. For all $i \in [\![1, r]\!]$, define $G_i = \sum_{k=1}^{r} a_{ik} F_k$. Since the F_i's are all in the ideal $I(V)$, then $G_i \in I(V)$. Moreover, for all $j \in [\![1, r]\!]$, we have that $G_i(P_j) = \sum_{k=1}^{r} a_{ik} F_k(P_j) = a_{ij}$.

1.18. Let I be an ideal in a ring R. If $a^n \in I$, $b^m \in I$, show that $(a+b)^{n+m} \in I$. Show that $\mathrm{Rad}(I)$ is an ideal. Show that any prime ideal is radical.

Solution By the Binomial Formula, we have that

$$(a+b)^{n+m} = \sum_{k=0}^{n+m} \binom{n+m}{k} a^k b^{n+m-k}.$$

When $k \in [\![0, n-1]\!]$, we have that the term of index k is in I because it can be written as $\binom{n+m}{k} a^k b^{n-k} \cdot b^m$ and b^m is in I. Similarly, if $k \in [\![n, n+m]\!]$, the term of index k can be written as $\binom{n+m}{k} a^{k-n} b^{n+m-k} \cdot a^n$, and $a^n \in I$, so it is also in I. All the terms in the sum above are in I so $(a+b)^{n+m}$ must be in I.

Since $\mathrm{Rad}(I) \supset I$ and $0 \in I$, then $0 \in \mathrm{Rad}(I)$. If $a, b \in \mathrm{Rad}(I)$, then there exist positive integers n and m such that $a^n, b^m \in I$. By the previous part, $(a+b)^{n+m} \in I$ which implies that $a + b \in \mathrm{Rad}(I)$. Thus, $\mathrm{Rad}(I)$ is closed under addition. Finally, if $a \in \mathrm{Rad}(I)$ and $r \in R$, then there is a positive integer n such that $a^n \in I$. Since $(ra)^n = r^n a^n \in I$, then $ra \in \mathrm{Rad}(I)$. Therefore, $\mathrm{Rad}(I)$ is an ideal.

Let I be a prime ideal, to prove that I is radical, it suffices to prove that $\mathrm{Rad}(I) \subset I$. Let $i \in \mathrm{Rad}(I)$, then $i^n \in I$ where $n \geq 1$ is the smallest positive integer for which $i^n \in I$. If $n = 1$, we are done. Suppose that $n > 1$, then we have that $i \cdot i^{n-1} \in I$. Since I is prime, then either i or i^{n-1} is in I. Since n is minimal, then both cases are impossible. Therefore, $\mathrm{Rad}(I) = I$ and so I is radical.

1.19. Show that $I = (X^2 + 1) \subset \mathbb{R}[X]$ is a radical (even a prime) ideal, but I is not the ideal of any set in $\mathbb{A}^1(\mathbb{R})$.

Solution Suppose that $p^n \in (X^2+1)$, then $X^1 + 2 \mid p^n$. Since X^2+1 is irreducible in $\mathbb{R}[X]$, then $X^2 + 1 \mid p$ which implies that $p \in (X^2 + 1)$. Therefore, $(X^2 + 1)$ is radical. Suppose that $(X^2 + 1) = I(X)$ for some $X \subset \mathbb{A}^1(\mathbb{R})$. If $X = \varnothing$, then $(X^2 + 1) = I(\varnothing) = \mathbb{R}[X]$ which is false since $X \notin (X^2 + 1)$. Hence, there is a $x \in \mathbb{A}^1(\mathbb{R})$ such that $x \in X$. Since $X^2 + 1 \in I(X)$, then $x^2 + 1 = (X^2 + 1)(x) = 0$ which is impossible since $X^2 + 1$ has no roots in $\mathbb{R}$. Therefore, $(X^2 + 1)$ is not the ideal of any set in $\mathbb{A}^1(\mathbb{R})$.

1.20. Show that for any ideal I in $k[X_1, ..., X_n]$, $V(I) = V(\mathrm{Rad}(I))$, and $\mathrm{Rad}(I) \subset I(V(I))$.

Solution Since $I \subset \mathrm{Rad}(I)$, then $V(\mathrm{Rad}(I)) \subset V(I)$. Conversely, let $P \in V(I)$ and $F \in \mathrm{Rad}(I)$, then $F^n \in I$ for some $n \geq 1$. It follows that $F^n(P) = 0$ and so $F(P)$. Thus, $P \in V(\mathrm{Rad}(I))$. Therefore, $V(I) = V(\mathrm{Rad}(I))$.

Now, from the fact that $S \subset I(V(S))$ for all subsets S of $k[X_1, ..., X_n]$, if we take $S = \mathrm{Rad}(I)$, then $\mathrm{Rad}(I) \subset I(V(\mathrm{Rad}(I))) = I(V(I))$ by the first part of the exercise.

1.21. Show that $I = (X_1 - a_1, ..., X_n - a_n) \subset k[X_1, ..., X_n]$ is a maximal ideal, and that the natural homomorphism from k to $k[X_1, ..., X_n]/I$ is an isomorphism.

Solution Let J be an ideal properly containing I, and let $F \in J \setminus I$. Write F as

$$F(X_1, ..., X_n) = F_k(X_1, ..., X_{n-1})X_n^k + \cdots + F_0(X_1, ..., X_{n-1}).$$

Since $F \in J$ and $F_k(X_1, ..., X_{n-1})(X_n - a_n)^k \in I \subset J$, then $F - F_k(X_1, ..., X_{n-1})(X_n - a_n)^k \in J$ where the resulting polynomial is now a polynomial in $k[X_1, ..., X_{n-1}][X_n]$ of degree $k-1$. By repeating this process, we can obtain a polynomial $G \in J$ which is a polynomial in $k[X_1, ..., X_{n-1}]$. Again, if we repeat the same procedure, we get a constant polynomial $H \in J$. This constant polynomial is nonzero because otherwise, we would get that F is the sum of all the polynomials in I that we subtracted and so F would be in I. Thus, $H \in J$ is a nonzero constant so $1 \in J$. It follows that $J = k[X_1, ..., X_n]$. Therefore, I is maximal.

Consider the natural homomorphism $\varphi : k \to k[X_1, ..., X_n]/I$ that maps each scalar to equivalence class of its respective constant polynomial. This is a well defined homomorphism because it is equal to the natural homomorphism from k to $k[X_1, ..., X_n]$ composed with the projection homomorphism from $k[X_1, ..., X_n]$ to $k[X_1, ..., X_n]/I$. This homomorphism is injective because it is a field homomorphism. This homomorphism is injective because in the first part of this exercice, we showed that every polynomial in $k[X_1, ..., X_n]$ is equivalent to a constant polynomial. Therefore, φ is an isomorphism.

1.4 The Hilbert Basis Theorem

1.22. Let I be an ideal in a ring R, $\pi : R \to R/I$ the natural homomorphism. (a) Show that for every ideal J' of R/I, $\pi^{-1}(J') = J$ is an ideal of R containg I, and for every ideal J of R containg I, $\pi(J) = J'$ is an ideal of R/I. This sets up a natural one-to-one correspondence between {ideals of R/I} and {ideals of R that contain I}. (b) Show that J' is a radical ideal if and only if J is radical. Similarly for prime and maximal ideals. (c) Show that J' is finitely generated if J is. Conclude that R/I is Noetherian if R is Noetherian. Any ring of the form $k[X_1, ..., X_n]/I$ is Noetherian.

Solution

(a) Let J' be an ideal of R/I and consider the set $J = \pi^{-1}(J')$. Since $\pi(0) = I = 0 \in J'$, then $0 \in \pi^{-1}(J') = J$. Let $a, b \in J$, then $\pi(a), \pi(b) \in J'$ which implies that $\pi(a+b) = \pi(a) + \pi(b) \in J'$. It follows that $a + b \in J$. Similarly, if $a \in J$ and $r \in R$, then $\pi(a) \in J'$ which implies that $\pi(ra) = \pi(r)\pi(a) \in J'$. Hence, $ra \in J$. Thus, J is an ideal of R. Finally, for all $i \in I$, we have that $\pi(i) = 0 \in J'$ so $i \in J$. Therefore, J is an ideal of R containing I.

Next, let J be an ideal of R that contains I, and define $J' = \pi(J)$. Since $0 \in J$ and $\pi(0) = 0$, then $0 \in J'$. Let $\pi(a), \pi(b) \in J'$, then $\pi(a)+\pi(b) = \pi(a+b) \in J'$ since $a + b \in J$. If $\pi(r) \in R/I$ and $\pi(a) \in J'$, then $\pi(r)\pi(a) = \pi(ra) \in J'$ since $ra \in J$. Therefore, J' is an ideal of R/I.

(b) Suppose that J' is radical and let $j^n \in J$, then $\pi(j)^n = \pi(j^n) \in J'$. Since J' is radical, then $\pi(j) \in J' = \pi(J)$ which implies that $j \in \pi^{-1}(\pi(J))$. Therefore, J is radical. Conversely, suppose that J is radical and let $\pi(j)^n \in J'$, then $\pi(j^n) \in J' = \pi(J)$. It follows that $j^n \in J$ which implies that $j \in J$. Thus, $\pi(j) \in J'$. Therefore, J' is radical.

Suppose that J' is prime and let $ab \in J$, then $\pi(a)\pi(b) = \pi(ab) \in J'$. By the primality of J', either $\pi(a)$ or $\pi(b)$ is in J'. Without loss of generality, if $\pi(a) \in J'$ then $a \in J$. It follows that J is prime. Conversely, if J is prime and $\pi(ab) = \pi(a)\pi(b) \in J'$, then $ab \in J$ which implies that $a \in J$ (wlog). It follows that $\pi(a) \in J'$. Therefore, J is prime.

Suppose that J' is maximal and let M be an ideal properly containing J, then its image $M' = \pi(M)$ is an ideal that contains J'. Let $m \in M \setminus J$, then $\pi(m) \in M' \setminus J'$ because if $\pi(m) \in J'$, then $m \in J$. Hence, M' properly contains J'. Since J' is maximal, then $M' = R/I$. By the bijection between the ideal classes, we must have $M = R$. Therefore, J is a maximal ideal. Conversely, suppose that J is maximal and let M' be an ideal that properly contains J'. Let $M = \pi^{-1}(M')$, then M is an ideal that contains J. Since M' contains J' properly, there must be an $m \in M$ such that $\pi(m) \in M' \setminus J'$. From this, it must be that $m \in M \setminus J$ and so M contains J properly. Since J is maximal, then $M = R$ and so $M' = R/I$. Therefore, J' is maximal.

(c) Suppose that J is finitely generated, then we can write $J = (j_1, ..., j_n)$. Let $\pi(j) \in J'$, then

$$\pi(j) = \pi\left(\sum_k r_k j_k\right) = \sum_k \pi(r_k)\pi(j_k) \in (\pi(r_1), \ldots, \pi(r_n)).$$

It follows that $J' \subset (\pi(r_1), \ldots, \pi(r_n))$. On the other hand, since the $\pi(j_k)$'s are in J', then $(\pi(r_1), \ldots, \pi(r_n)) \subset J'$. Therefore, J' is finitely generated.

Now, suppose that R is Noetherian and let J' be an ideal of R/I, then $J' = \pi(J)$ where J is an ideal of R containing I. Since R is Noetherian, then J is finitely generated and so J' is also finitely generated. Therefore, R/I is Noetherian.

1.5 Irreducible Components of an Algebraic Set

1.23. Give an example of a collection $\mathscr{S}$ of ideals in a Noetherian ring such that no maximal member of $\mathscr{S}$ is a maximal ideal.

Solution Take the ring to be $\mathbb{Z}$ and take the collection $\mathscr{S} = \{(4)\}$ containing only one element. It is clear that (4) is a maximal element of $\mathscr{S}$ even though it is not a maximal ideal of $\mathbb{Z}$.

1.24. Show that every proper ideal in a Noetherian ring is contained in a maximal ideal. (*Hint:* If I is the ideal, apply the lemma to $\{$proper ideals that contain $I\}$.)

Solution Let I be a proper ideal and let $\mathscr{S}$ be the collection of proper ideals that contain I. By the lemma, $\mathscr{S}$ must contain a maximal element J that contain I. If J is not a maximal ideal, then it must be itself contained in a proper ideal of the ring. However, this new proper ideal would be in $\mathscr{S}$ and would contradict the maximality of J in $\mathscr{S}$. Therefore, J is a maximal ideal that contain I.

1.25. (a) Show that $V(Y - X^2) \subset \mathbb{A}^2(\mathbb{C})$ is irreducible; in fact, $I(V(Y - X^2)) = (Y - X^2)$. (b) Decompose $V(Y^4 - X^2, Y^4 - X^2Y^2 + XY^2 - X^3) \subset \mathbb{A}^2(\mathbb{C})$ into irreducible components.

Solution

(a) Let's show that $I(V(Y - X^2))$ is a prime ideal. First, notice that $I(V(Y - X^2)) = (Y - X^2)$. Since $Y - X^2$ is irreducible in $\mathbb{C}[X, Y]$, then $(Y - X^2)$ is prime. Therefore, $V(Y - X^2)$ is irreducible.

(b) First, notice that $Y^4 - X^2 = (Y^2 - X)(Y^2 + X)$ and $Y^4 - X^2Y^2 + XY^2 - X^3 = (Y^2 - X^2)(Y^2 + X) = (Y - X)(Y + X)(Y^2 + X)$. Let's prove that our algebraic set V is equal to $V(Y^2 + X) \cup \{(1, 1)\} \cup \{(1, -1)\}$. First, if $(x, y) \in V$, then $y^4 = x^2$ and $(y - x)(y + x)(y^2 + x) = 0$. From the second equation, we have the following cases: if $y = x$, then either $x = y = 0$ or $y^4 = x^2$ implies that $y^2 = 1$, and hence, that $x = y = \pm 1$; if $y = -x$, then either $x = y =$ or the same considerations as before gives us that $y = \pm 1$, and hence, $x = \mp 1$, $y = \pm 1$; finally, if $y^2 = -x$, then $y^2 + x = 0$. The outcome of all of these cases show that $(x, y) \in V(Y^2 + X) \cup \{(1, 1)\} \cup \{(1, -1)\}$. The converse is even easier to prove, if $y^2 + x = 0$, then (x, y) satisfy both polynomials describing V and so it is in V; if $x = 1$ and $y = \pm 1$, then it clearly satisfies both polynomials again and so it is in V. Therefore, $V = V(Y^2 + X) \cup \{(1, 1)\} \cup \{(1, -1)\}$. The three sets are clearly irreducible algebraic sets so we are done.

1.26. Show that $F = Y^2 + X^2(X - 1)^2 \in \mathbb{R}[X, Y]$ is an irreducible polynomial, but $V(F)$ is reducible.

Solution By contradiction, suppose that there exist two polynomials $p, q \in \mathbb{R}[X, Y]$ such that $y^2 + x^2(x - 1)^2 = p(x, y)q(x, y)$, then in particular, if we fix any $x \in \mathbb{R} \setminus \{0, 1\}$, we can view this equation as an equation where y is the only variable. But this would mean that the polynomial $y^2 + x^2(x - 1)^2$ is reducible in $\mathbb{R}[y]$ which is false. Therefore, the polynomial F is irreducible.

To show that $V(F)$ is reducible, notice that the only solutions to the equation $Y^2 + X^2(X - 1)^2 = 0$ are $(0, 0)$ and $(1, 0)$. Therefore, $V(F) = \{(0, 0)\} \cup \{(1, 0)\}$.

1.27. Let V, W be algebraic sets in $\mathbb{A}^n(k)$, with $V \subset W$. Show that each irreducible component of V is contained in some irreducible component of W.

Solution First, write $V = \bigcup_i V_i$ and $W = \bigcup_j W_j$ where the unions are finite. Since $V \subset W$, then for each i, we have $V_i = \bigcup_j (W_j \cap V_i)$. Since V_i is irreducible, then

we must have that $V_i = W_j \cap V_i \subset W_j$ for some j. Therefore, every irreducible component of V is contained in an irreducible component of W.

1.28. If $V = V_1 \cup \cdots \cup V_r$ is the decomposition of an algebraic set into irreducible components, show that $V_i \not\subset \bigcup_{i \neq j} V_j$.

Solution First, recall that $V_i \not\subset V_j$ for all $i \neq j$. Next, suppose that $V_i \subset \bigcup_{i \neq j} V_j$, then $V_i = \bigcup_{i \neq j}(V_j \cap V_i)$. But since V_i is irreducible, then $V_i = V_j \cap V_i \subset V_j$ for some $i \neq j$. This contradiction proves that the original claim holds.

1.29. Show that $\mathbb{A}^n(k)$ is irreducible if k is infinite.

Solution Simply notice that when k is infinite, we have $I(\mathbb{A}^n(k)) = \{0\}$ and $\{0\}$ is a prime ideal. Therefore, $\mathbb{A}^n(k)$ is irreducible.

1.6 Algebraic Subsets of the Plane

1.30. Let $k = \mathbb{R}$. (a) Show that $I(V(X^2 + Y^2 + 1)) = (1)$. (b) Show that every algebraic subset of $\mathbb{A}^2(\mathbb{R})$ is equal to $V(F)$ for some $F \in \mathbb{R}[X, Y]$.

Solution

(a) Since $X^2 + Y^2 + 1$ has no roots in $\mathbb{A}^2(\mathbb{R})$, then

$$I(V(X^2 + Y^2 + 1)) = I(\varnothing) = \mathbb{A}^2(\mathbb{R}) = (1).$$

(b) Let V be an algebraic subset of $\mathbb{A}^2(\mathbb{R})$. If $V = \mathbb{A}^2(\mathbb{R})$, then $V = V(0)$; if $V = \varnothing$, then $V = V(1)$; if $V = \{(a, b)\}$, then we can write V as $V((X-a)^2+(Y-b)^2)$ because $(X-a)^2+(Y-b)^2 = 0$ if and only if $X = a$ and $Y = b$; if V is any other algebraic set, then we already know that $V = V(F)$ for some $F \in \mathbb{R}[X, Y]$ using Corollary 2.

1.31. (a) Find the irreducible components of $V(Y^2-XY-X^2Y+X^3)$ in $\mathbb{A}^2(\mathbb{R})$, and also in $\mathbb{A}^2(\mathbb{C})$. (b) Do the same for $V(Y^2-X(X-1)^2)$, and for $V(X^3+X-X^2Y-Y)$.

Solution

(a) Notice that $Y^2 - XY - X^2Y + X^3 = (Y - X^2)(Y - X)$, then $V(Y^2 - XY - X^2Y + X^3) = V(Y - X^2) \cup V(Y - X)$. This decomposition is the irreducible decomposition in $\mathbb{C}[X, Y]$ because both $Y - X^2$ and $Y - X$ are irreducible, and using Corollary 3. This is also the irreducible decomposition in $\mathbb{A}^2(\mathbb{C})$ because both $Y - X^2$ and $Y - X$ are irreducible in $\mathbb{R}[X, Y]$ and both have an infinite associated algebraic set.

(b) The first polynomial is irreducible and has infinitely many solutions in $\mathbb{A}^2(\mathbb{R})$, hence, its associated algebraic set is irreducible. The second polynomial has the same factorization into irreducible polynomials in $\mathbb{A}^2(\mathbb{R})$ and in $\mathbb{A}^2(\mathbb{C})$ so again, the factorization is the same in the two cases.

1.7 Hilbert's Nullstellensatz

1.32. Show that both theorems and all of the corollaries are false if k is not algebraically closed.

Solution For the Weak-Nullstellensatz, take $k = \mathbb{R}$, $n = 1$, $I = (X^2 + 1)$, then $V(I) = \varnothing$ since $X^2 + 1$ has no roots in $\mathbb{R}$. The same set up disproves the Hilbert's Nullstellensatz, as well as Corollary 1 and 2. For the third corollary, it suffices to look at $V(Y^2 + X^2(X-1)^2)$ in $\mathbb{R}$ and see that even though the polynomial is irreducible, the algebraic set is reducible into $\{(0,0)\}$ and $\{(1,0)\}$. Similarly, if we take $I = (X^2+Y^2) \subset \mathbb{R}[X,Y]$, then $V(I) = \{(0,0)\}$ is finite but $\mathbb{R}[X,Y]/(X^2+Y^2)$ is infinite dimensional. This last example shows that corollary 4 doesn't hold if k is not algebraically closed.

1.33. (a) Decompose $V(X^2 + Y^2 - 1, X^2 - Z^2 - 1) \subset \mathbb{A}^3(\mathbb{C})$ into irreducible components. (b) Let $V = \{(t, t^2, t^3) \in \mathbb{A}^3(\mathbb{C}) | t \in \mathbb{C}\}$. Find $I(V)$, and show that V is irreducible.

Solution

(a) First, notice that $(x^2 + y^2 - 1, x^2 - z^2 - 1) = (x^2 + y^2 - 1, (y + iz)(y - iz))$. It follows that

$$V(x^2 + y^2 - 1, x^2 - z^2 - 1) = V(x^2 + y^2 - 1, y + iz) \cup V(x^2 + y^2 - 1, y - iz).$$

Since

$$\begin{aligned}\mathbb{C}[x,y,z]/(x^2 + y^2 - 1, y + iz) &\cong (\mathbb{C}[x,y,z]/(y + iz))/(x^2 + y^2 - 1)\\ &\cong \mathbb{C}[x,y]/(x^2 + y^2 - 1)\end{aligned}$$

is an integral domain, then the ideal $(x^2 + y^2 - 1, y + iz)$ is prime, and hence, $V(x^2 + y^2 - 1, y + iz)$ is irreducible. Similarly, the algebraic set $V(x^2 + y^2 - 1, y - iz)$ is irreducible as well. Therefore, we have found the decomposition of the algebraic set $V(x^2 + y^2 - 1, x^2 - z^2 - 1)$ into irreducible algebraic sets.

(b) First, notice that $V = V(Y - X^2, Z - X^3)$. Hence, if we let $I = (Y - X^2, Z - X^3)$ and notice that $\mathbb{C}[X,Y,Z]/I \cong \mathbb{C}[X]$ is an integral domain, then we get that I is a prime ideal, and hence, $I(V) = I(V(I)) = I = (Y - X^2, Z - X^3)$ by the Nullstellensatz.

1.34. Let R be a UFD. (a) Show that a monic polynomial of degree two or three in $R[X]$ is irreducible if and only if it has no roots in R. (b) The polynomial $X^2 - a \in R[X]$ is irreducible if and only if a is not a square in R.

Solution

(a) First, notice that if a polynomial of degree 2 or 3 is reducible, then it must be divisible by a polynomial of degree 1. Moreover, in the decomposition of the polynomial into two smaller (in the sense of the degree) polynomials, the leading coefficients of each smaller polynomial must be inverses of each other

since the original polynomial is monic. In that case, we can multiply the two polynomials by constants such that the degree one polynomial dividing the original polynomial is monic as well. It follows that a reducible, monic, degree 2 or 3 polynomial must have a root since it is divisible by a polynomial which has a root. The converse is easy.

(b) This follows from part (a) using the fact that $X^2 - a$ has a root if and only if a has a root in R.

1.35. Show that $V(Y^2 - X(X-1)(X-\lambda)) \subset \mathbb{A}^2(k)$ is an irreducible curve for any algebraically closed field k, and any $\lambda \in k$.

Solution Let's show that the polynomial $y^2 - x(x-1)(x-\lambda)$ is irreducible. By contradiction, let $p(x,y)$ and $q(x,y)$ be two nontrivial polynomials such that $p(x,y)q(x,y) = y^2 - x(x-1)(x-\lambda)$, then the highest degree of y in p_1 and p_2 must be either 0 and 2, or 1 in both. However, the first case would imply that $y^2 - x(x-1)(x-\lambda)$ is factorizable by a polynomial in x, which is not the case. Thus, we must have $p(x,y) = p_1(x)y + p_0(x)$ and $q(x,y) = q_1(x)y + q_0(x)$ for some one variable polynomials p_1, p_0, q_1, q_0. In that case,

$$\begin{aligned} &y^2 - x(x-1)(x-\lambda) \\ &\quad = p_1(x)q_1(x)y^2 + (p_1(x)q_0(x) + p_0(x)q_1(x))y + p_0(x)q_0(x) \end{aligned}$$

which implies that $p_1(x)q_1(x) = 1$, $p_1(x)q_0(x) + p_0(x)q_1(x) = 0$ and $p_0(x)q_0(x) = x(x-1)(x-\lambda)$. Since $p_1(x)q_1(x) = 1$, then $p_1(x) = q_1(x) = 1$ without loss of generality. It follows that $q_0(x) = -p_0(x)$, and hence, $-p_0(x)^2 = x(x-1)(x-\lambda)$. But $x(x-1)(x-\lambda)$ is a polynomial of degree 3 while $-p_0(x)^2$ must have an even degree. Therefore, by contradiction, $y^2 - x(x-1)(x-\lambda)$ is irreducible, implying that $V(Y^2 - X(X-1)(X-\lambda))$ is irreducible.

1.36. Let $I = (Y^2 - X^2, Y^2 + X^2) \subset \mathbb{C}[X,Y]$. Find $V(I)$ and $\dim_{\mathbb{C}}(\mathbb{C}[X,Y]/I)$.

Solution Notice that $I = (Y^2 - X^2, Y^2 + X^2) = (X^2, Y^2)$. Hence, $V(I) = V(X^2, Y^2) = \{(0,0)\}$. Since

$$\mathbb{C}[X,Y]/(X^2,Y^2) \cong \{c_0 + c_1X + c_1Y + c_2XY : c_i \in \mathbb{C}\},$$

then $\dim_{\mathbb{C}}(\mathbb{C}[X,Y]/I) = 4$.

1.37. Let k be any field, $F \in K[X]$ a polynomial of degree $n > 0$. Show that the residues of $\overline{1}, \overline{X}, ..., \overline{X}^{n-1}$ form a basis of $K[X]/(F)$ over K.

Solution First, it is clear that the elements $\overline{1}, \overline{X}, ..., \overline{X}^{n-1}$ span the elements in $K[X]/(F)$ since every polynomial in $K[X]$ is equivalent to a polynomial of degree at most $n-1$ modulo F. Next, let's show that n elements are linearly independent: let $c_0, c_1, ..., c_{n-1} \in K$ such that

$$c_0\overline{1} + c_1\overline{X} + \cdots + c_{n-1}\overline{X}^{n-1} = \overline{0},$$

in $K[X]/(F)$, then equivalently:

$$c_0 + c_1X + \cdots + c_{n-1}X^{n-1} = p(X)F(X)$$

for some polynomial $p \in K[X]$. The degree of the left hand side is necessarily less than or equal to $n-1$; however, the degree of the right hand side is either 0 or strictly greater than $n-1$. For the equality to hold, it must be that the left hand side has degree 0, and hence, that $c_i = 0$ for all i.

Therefore, the elements $\overline{1}, \overline{X}, \ldots, \overline{X}^{n-1}$ are linearly independent, and hence, they form a basis of $K[X]/(F)$.

1.38. Let $R = k[X_1, ..., X_n]$, k algebraically closed, $V = V(I)$. Show that there is a natural one-to-one correspondence between algebraic subsets of V and radical ideals in $k[X_1, ..., X_n]/I$, and that irreducible algebraic sets (resp. points) correspond to prime ideals (resp. maximal ideals). (See Problem 1.22.)

Solution We already know that there is a one-to-one correspondence between the radical ideals of $k[X_1, ..., X_n]/I$ and the radical ideals of $k[X_1, ..., X_n]$ containing I. By Nullstellensatz, we know that there is a one-to-one correspondence between the radical ideals of $k[X_1, ..., X_n]$ and the algebraic sets. Since $V(J) \supset V(I)$ if and only if $J \subset I$, then in particular, there is a one-to-one correspondence between the radical ideals of $k[X_1, ..., X_n]$ containing I and the algebraic sets contained in $V(I)$. Therefore, there is a one-to-one correspondence between the radical ideals of $k[X_1, ..., X_n]/I$ and the algebraic sets contained in $V(I)$. Since an ideal of R/I is prime if and only if its associated ideal is prime in R, then an ideal is prime in R/I if and only if the associated algebraic set conatined in $V(I)$ is irreducible (from the equivalence between prime and irreducible set). The logic is the same for the maximal ideal case.

1.39. (a) Let R be a UFD, and let $P = (t)$ be a principal, proper, prime ideal. Show that there is no prime ideal Q such that $0 \subset Q \subset P$, $Q \neq 0$, $Q \neq P$. (b) Let $V = V(F)$ be an irreducible hypersurface in $\mathbb{A}^n$. Show that there is no irreducible algebraic set W such that $V \subset W \subset \mathbb{A}^n$, $W \neq V$, $W \neq \mathbb{A}^n$.

Solution

(a) Suppose that there is an ideal Q such that $0 \subset Q \subset P$, $Q \neq 0$, $Q \neq P$. Since Q is nonzero, then there is a nonzero element $q \in Q$. Since R is a UFD, then there is a maximal number n such that t divides q since t is irreducible. Since $Q \subset P$, then $q = k_1t$ for some $k \in R$. But Q is prime so $k_1t \in Q$ implies that $k_1 \in Q$ or $t \in Q$. But t cannot be in Q since otherwise, $P \subset Q$, a contradiction. Hence, $k_1 \in Q$ which implies that $k_1 = k_2t$. Applying this argument as much as we want implies that for all i, there is an integer k_i such that $q = k_it^i$. In particular, when $i = n+1$, we get that t^{n+1} divides q. But this contradicts the fact that n is maximal. Therefore, such a prime ideal Q cannot exist.

(b) If we consider the irreducible divisors of F, then the fact that $V(F)$ is irreducible implies that $F = F_0^n$ where F_0 is irreducible. It follows that $I(V) = (F_0)$ is principal, proper and prime. Hence, applying part (a) to $R = k[X_1, ..., X_n]$ and $t = F_0$ gives us that there is no prime ideal J such that $0 \subsetneq J \subsetneq I(V)$.

Since there is a one-to-one correspondence between prime ideals and irreducible algebraic sets, the last statement is equivalent to the fact that there is no irreducible algebraic set W such that $V \subsetneq W \subsetneq \mathbb{A}^n$.

1.40. Let $I = (X^2 - Y^3, Y^2 - Z^3) \subset k[X, Y, Z]$. Define $\alpha : k[X, Y, Z] \to k[T]$ by $\alpha(X) = T^9$, $\alpha(Y) = T^6$, $\alpha(Z) = T^4$. (a) Show that every element of $k[X, Y, Z]/I$ is the residue of an element $A + XB + YC + XYD$ for some $A, B, C, D \in k[Z]$. (b) If $F = A + XB + YC + XYD$, $A, B, C, D \in k[Z]$, and $\alpha(F) = 0$, compare like powers of T to conclude that $F = 0$. (c) Show that $\ker(\alpha) = I$, so I is prime, $V(I)$ is irreducible, and $I(V(I)) = I$.

Solution

(a) Simply notice that for polynomial F in $k[X, Y, Z]$, this polynomial is equivalent to a polynomial $PX+Q$, $P, Q \in k[Y, Z]$, modulo X^2-Y^3, and every polynomial $H \in k[Y, Z]$ is equivalent to $RY + S$, $R, S \in k[Z]$, modulo $Y^2 - Z^3$. It follows that every element in $k[X, Y, Z]$ is equivalent to a polynomial of the form $A + XB + YC + XYD$ for some polynomials $A, B, C, D \in k[Z]$ modulo I. It follows that every element in $k[X, Y, Z]/I$ is the residue of such a polynomial.

(b) If $F = A + XB + YC + XYD$ with $\alpha(F) = 0$, then

$$0 = \alpha(A) + T^9\alpha(B) + T^6\alpha(C) + T^{15}\alpha(D).$$

Since $\alpha(Z) = T^4$, then the powers of T in $\alpha(A)$ are all multiples of 4. Similarly, the powers of T in $T^9\alpha(B)$ are all of the form $4k + 1$, the powers of T in $T^6\alpha(C)$ are all of the form $4k + 2$ and the powers of T in $T^{15}\alpha(D)$ are all of the form $4k + 3$. Therefore, the only way the sum can be equal to zero is when all the coefficients of the polynomials A, B, C, D are zero, and hence, that $A = B = C = D = 0$, or finally, when $F = 0$.

(c) Since $\alpha(X^2 - Y^3) = \alpha(Y^2 - Z^3) = 0$, then $I \subset \ker(\alpha)$. Next, by part(a), we know that for an arbitrary F, we have that $F = A + XB + YC + XYD + E$ where $A, B, C, D \in k[Z]$ and $E \in I$. Hence, $\alpha(F) = 0$ implies that

$$\alpha(A + XB + YC + XYD) + \alpha(E) = 0.$$

But $I \subset \ker(\alpha)$ so $\alpha(E) = 0$. Hence,

$$\alpha(A + XB + YC + XYD) = 0$$

which implies that $A + XB + YC + XYD = 0$ by part (b). Thus, $F = E \in I$ which proves that $\ker(\alpha) \subset I$. Therefore, $\ker(\alpha) = I$.

To show that I is prime, let $AB \in I = \ker(\alpha)$ which implies that $\alpha(AB) = 0$. But $\alpha(AB) = \alpha(A)\alpha(B)$ so either $\alpha(A)$ or $\alpha(B)$ is zero. Equivalently, one of A or B must be in $\ker(\alpha) = I$. Therefore, I is prime. By the Nullstellensatz, it follows that $V(I)$ is irreducible, and $I(V(I)) = I$.

1.8 Modules; Finiteness Conditions

1.41. If S is module-finite over R, then S is ring-finite over R.

Solution If S is module-finite over R, then there are elements $s_1, ..., s_n \in S$ such that every element in S is of the form $r_1 s_1 + \cdots + r_n s_n$ with $r_i \in R$. Consider now the ring $R[s_1, ..., s_n]$, the smallest ring containing R and the elements $s_1, ..., s_n$. Clearly, this ring is contained in S since S is a ring conatining R and the elements $s_1, ..., s_n$. Conversely, if $r_1 s_1 + \cdots + r_n s_n \in S$, then $r_1 s_1 + \cdots + r_n s_n \in R[s_1, ..., s_n]$ because $R[s_1, ..., s_n]$ is a ring, so it contains all the elements $r_i s_i$, and also their sum. Therefore, $S = R[s_1, ..., s_n]$, so S is ring-finite over R.

1.42. Show that $S = R[X]$ (the ring of polynomials in one variable) is ring-finite over R, but not module-finite.

Solution It is clear that the ring S is ring-finte since it is equal to the smallest ring containing R and X. Next, suppose that S is module-finite, then there exist polynomials $p_1, ..., p_n$ such that every polynomial can be written as an R-linear combination of $p_1, ..., p_n$. However, if we let N be the maximal degree of the p_i's, then any linear combination of the p_i's will have a degree at most N. It follows that not all polynomials in S can be written as an R-linear combination of the p_i's. Therefore, S is not module-finite.

1.43. If L is ring-finite over K (K, L fields) then L is a finitely generated field extension of K.

Solution If L is ring-finite over K, then $L = K[v_1, ..., v_n]$ for some $v_1, ..., v_n \in L$. But since L is a field, then the quotient field of $K[v_1, ..., v_n]$ is itself, i.e., $K[v_1, ..., v_n] = K(v_1, ..., v_n)$. Therefore $L = K(v_1, ..., v_n)$, which proves that L is a finitely generated field extension of K.

1.44. Show that $L = K(X)$ (the field of rational functions in one variable) is a finitely generated field extension of K, but L is not ring-finite over K. (*Hint:* If L were ring-finite over K, a common denominator of ring generators would be an element $b \in K[X]$ such that for all $z \in L$, $b^n z \in K[X]$ for some n; but let $z = 1/c$, where c doesn't divide b (Problem 1.5).)

Solution It is clear that L is a finitely generated field extension of K since it can be written as the $K(X)$. Next, suppose that $L = K[p_1/q_1, ..., p_n/q_n]$ for some $p_i/q_i \in L$, then every rational function in one variable can be written as a polynomial in p_i/q_i. But notice that for any polynomial in p_i/q_i, there exists an integer $N \geq 1$ such that multiplying the polynomial by $(q_1 \cdots q_n)^N$ gives a polynomial in X. Now, consider the rational function $1/(q_1 \cdots q_n + 1) \in L$. For all $N \geq 1$, $(q_1 \cdots q_n)^N/(q_1 \cdots q_n + 1)$ is a not a polynomial in X because the numerator is not divisible by the denominator. Thus, by contradiction, L is not ring-finite.

1.45. Let R be a subring of S, S a subring of T.

(a) If $S = \sum Rv_i$, $T = \sum Sw_j$, show that $T = \sum Rv_i w_j$.

(b) If $S = R[v_1, ..., v_n]$, $T = S[w_1, ..., w_m]$, show that $T = R[v_1, ..., v_n, w_1, ..., w_m]$.

(c) If R, S, T are fields, and $S = R(v_1, ..., v_n)$, $T = S(w_1, ..., w_m)$, show that $T = R(v_1, ..., v_n, w_1, ..., w_m)$.

Solution

(a) Let $\sum s_j w_j$ be an arbitrary element of T, then each s_j can be written as $\sum_i r_{ij} v_i$. Hence:

$$\sum_j s_j w_j = \sum_j \left(\sum_i r_{ij} v_i \right) w_j = \sum_j \sum_i r_{ij} v_i w_j.$$

Therefore, $T = \sum R v_i w_j$ because every element in T can be written as an R-linear combination of $v_i w_j$.

(b) Let $p(w_1, ..., w_m)$ be an arbitrary element in T with $p \in S[X_1, ..., X_m]$, then $p(w_1, ..., w_m)$ is a succession of additions and multiplications of elements in S and $w_1, ..., w_m$. But since every element s in S can be written as a polynomial $p_s(v_1, ..., v_n)$ with $p_s \in R[X_1, ..., X_n]$, then $p(w_1, ..., w_m)$ is an addition and multiplication of polynomials in $v_1, ..., v_n$ and $w_1, ..., w_m$. Since additions and multiplications of polynomials in $v_1, ..., v_n, w_1, ..., w_m$ is still a polynomial in $v_1, ..., v_n, w_1, ..., w_m$, then every element of T can be written as a polynomial in $v_1, ..., v_n, w_1, ..., w_m$. Therefore, $T = R[v_1, ..., v_n, w_1, ..., w_m]$.

(c) Since T contains S and the elements $w_1, ..., w_m$, and S contains R and the elements $v_1, ..., v_n$, then T is a field containing R and the elements $v_1, ..., v_n$, $w_1, ..., w_m$. It follows that $R(v_1, ..., v_n, w_1, ..., w_m) \subseteq T$. Conversely, since $R(v_1, ..., v_n) \subseteq R(v_1, ..., v_n, w_1, ..., w_m)$, then $S \subseteq R(v_1, ..., v_n, w_1, ..., w_m)$. It follows that $T = S(w_1, ..., w_m) \subseteq R(v_1, ..., v_n, w_1, ..., w_m)$. Therefore, $T = R(v_1, ..., v_n, w_1, ..., w_m)$.

1.9 Integral Elements

1.46. Let R be a subring of S, S a subring of (a domain) T. If S is integral over R, and T is integral over S, show that T is integral over R. (*Hint:* Let $z \in T$, so we have $z^n + a_1 z^{n-1} + ... + a_n = 0$, $a_i \in S$. Then $R[a_1, ..., a_n, z]$ is module-finite over R.)

Solution Let $z \in T$, since T is integral over S, then there exists elements $a_1, ..., a_n \in S$ such that $z^n + a_1 z^{n-1} + ... + a_n = 0$. It follows that z is integral over $R[a_1, ..., a_n]$, so the ring $R[a_1, ..., a_n, z]$ is module-finite. Since $R[v]$ is a subring of $R[a_1, ..., a_n, z]$, then z is integral over R by Proposition 3. Therefore, T is integral over R.

1.47. Suppose (a domain) S is ring-finite over R. Show that S is module-finite over R if and only if S is integral over R.

Solution If S is module-finite over R, then for all $z \in S$, we have that $R[z]$ is a subring of S, which is module-finite over R, so z is integral over R. Therefore, S is integral over R.

Conversely, suppose that S is integral over R. Since S is ring-finite over R, then $S = R[v_1, ..., v_n]$ for some $v_1, ..., v_n \in S$. For all i, there exists an integer n_i such that every power v_i^m with $m > n_i$ can be written in terms of v_i, v_i^2, ..., $v_i^{n_i}$ (because v_i solves a polynomial equation). Since $S = R[v_1, ..., v_n]$, then every element in S can be written as $p(v_1, ..., v_n)$ where $p(x_1, ..., x_n)$ is a polynomial in $R[x_1, ..., x_n]$. Since every polynomial is a sum of monomials of the form $\prod_i v_i^{m_i}$, then rewriting each higher powers of v_i in terms of the lower powers gives us that $p(v_1, ..., v_n)$ is a sum of monomials of the form $\prod_i v_i^{m_i}$ with the restriction $m_i \leq n_i$. Thus, every element of S can be written as a linear combination of these monomials, which there are finitely many. Thus, S is generated as a module by these monomials. Therefore, S is module-finite.

1.48. Let L be a field, k an algebraically closed subfield of L. (a) Show that any element of L that is algebraic over k is already in k. (b) An algebraically closed field has no module-finite field extensions except itself.

Solution

(a) Let $z \in L$ be algebraic over k, then there is a (monic) polynomial $p(x) \in k[x]$ such that $p(z) = 0$. Since k is algebraically closed, then $p(x) = (x - a_1) \cdots (x - a_k)$ for some $a_1, ..., a_n \in k$. But since $(z - a_1) \cdots (z - a_k) = 0$, then $z = a_i$ for some i, and hence, $z \in k$.

(b) Let K be a module-finite extension of k and let $z \in K$, then $k[z]$ is contained in a module-finite ring over k, so z must be integral over k. Hence, z is algebraic over k, which implies that $z \in k$ by part (a). Therefore, $K = k$.

1.49. Let K be a field, $L = K(X)$ the field of rational functions in one variable over K. (a) Show that any element of L that is integral over $K[X]$ is already in $K[X]$. (*Hint:* If $z^n + a_1 z^{n-1} + \cdots = 0$, write $z = F/G$, F, G relatively prime. Then $F^n + a_1 F^{n-1} G + \cdots = 0$, so G divides F.) (b) Show that there is no nonzero element $F \in K[X]$ such that for every $z \in L$, $F^n z$ is integral over $K[X]$ for some $n > 0$. (*Hint:* See Problem 1.44.)

Solution

(a) Let $p/q \in K(X)$ be integral over $K[X]$, then there exist polynomials $a_1, ..., a_n \in K[X]$ such that

$$\frac{p^n}{q^n} + a_1 \frac{p^{n-1}}{q^{n-1}} + \cdots + a_{n-1} \frac{p}{q} + a_n = 0.$$

Multiplying both sides by q^n and isolating p^n gives us

$$p^n = -a_1 p^{n-1} q - \cdots - a_{n-1} p q^{n-1} - a_n q^n.$$

Now, let b be an irreducible divisor of q, then b divides the right hand side, and hence, the left hand side which is equal to p^n. Since b is irreducible, then

b divides p. But this shows that every irreducible factor of q divides p, which directly implies that q divides. Hence, $p/q \in K[X]$.

(b) Such an element cannot exist because it would mean that for every $p/q \in L$, there is a n such that $F^n p/q \in K[X]$. If $p = 1$ and $q = F + 1$, then $F^n/(F+1) \notin K[X]$ for all n, contradicting the existence of F.

1.50. Let K be a subfield of a field L. (a) Show that the set of elements of L that are algebraic over K is a subfield of L containing K. (*Hint:* If $v^n + a_1 v^{n-1} + \cdots + a_n = 0$, and $a_n \neq 0$, then $v(v^{n-1} + \dots) = -a_n$.) (b) Suppose L is module-finite over K, and $K \subset R \subset L$. Show that R is a field.

Solution

(a) An element of L is algebraic over K if and only if it is integral over K. Hence, the subset of elements of L algebraic over K form a subring of L. Hence, it suffices to show that if an element is algebraic over K, then its inverse is algebraic over K. Let $z \in L$ be algebraic over K, then $a_n z^n + \cdots + a_0 = 0$ for some $a_i \in K$ such that $a_0 \neq 0$. It follows that $a_0(z^{-1})^n + \cdots + a_n = 0$ which proves that z^{-1} is algebraic pver K. Therefore, the elements that are algebraic over K form a subfield of L which contains K (because every element of K is algebraic over K).

(b) There must be a mistake in the statement of this question because it is clearly false.

1.10 Field Extensions

1.51. Let K be a field $F \in K[X]$ an irreducible monic polynomial of degree $n > 0$. (a) Show that $L = K[X]/(F)$ is a field, and if x is the residue of X in L, then $F(x) = 0$. (b) Suppose L' is a field extension of K, $y \in L'$ such that $F(y) = 0$. Show that the homomorphism from $K[X]$ to L' that takes X to y induces an isomorphism of L with $K(y)$. (c) With L', y as in (b), suppose $G \in K[X]$ and $G(y) = 0$. Show that F divides G. (d) Show that $F = (X - x)F_1$, $F_1 \in L[X]$.

Solution

(a) By Problem 1.3, the ideal (F) is maximal so $K[X]/(F)$ is a field. In this field, the residue of $F(X)$ is 0. Since $F(x)$ is equal to the residue of $F(X)$, then $F(x) = 0$.

(b) The image of the homomorphism $\phi : K[X] \to L'$ mapping $p(X)$ to $p(y)$ is $K[y]$ by definition. Hence, we have the isomorphism $K[y] \cong K[X]/\ker\phi$. Since $K[X]$ is a PID, then $\ker\phi = (f)$ for some $f \in K[X]$. Since $\phi(F) = F(y) = 0$, then $F \in \ker\phi = (f)$. It follows that f divides F, and so by irreducibility of F, f is either 1 or F. If $f = 1$, then $\ker\phi = k[X]$ which is a contradiction, so $f = F$. Hence, $K[y] \cong K[X]/(F) \cong L$. Since L is a field, then $K[y] = K(y)$. Therefore, $L \cong K(y)$.

(c) If $G(y) = 0$, then $G \in \ker \phi = (F)$. It follows that F divides G.

(d) Since x is a root F, then viewing F as a polynomial in $L[X]$ gives us that $F = (X - x)F_1$ where $F_1 \in L[X]$.

1.52. Let K be a field, $F \in K[X]$. Show that there is a field L containing K such that $F = \prod_{i=1}^{n}(X - x_i) \in L[X]$. (*Hint:* Use Problem 1.51(d) and induction on the degree.) L is called a splitting field of F.

Solution When $n = 1$, it is trivial because it suffices to take $L = K$. When $n > 1$, suppose that the statement holds for all polynomials of degree strictly smaller than n. By Problem 1.51(d), we can construct a field L_0 that contains K and where $F = (X - a_n)F_1$ where $F_1 \in L[X]$. By induction, let L be a field containing L_0 such that $F_1 = (X - a_1) \cdots (X - a_{n-1})$, then L is a field containing K such that $F = (X - a_1) \cdots (X - a_n)$.

1.53. Suppose K is a field of characteristic zero, F an irreducible monic polynomial in $K[X]$ of degree $n > 0$. Let L be a splitting field of F, so $F = \prod_{i=1}^{n}(X - x_i)$, $x_i \in L$. Show that the x_i are distinct. (*Hint:* Apply Problem 1.51(c) to $G = F_X$; if $(X - x)^2$ divides F, then $G(x) = 0$.)

Solution Let x be a root of F in L and suppose that $(X - x)^2$ divides F, then $(X - x)$ divides $F_X \in K[X]$. Hence, $G(x) = 0$, so F divides F_X. But this is impossible since F_X has a degree smaller than F. Thus, $(X - x)^2$ doesn't divide F, and so the x_i are distinct.

1.54. Let R be a domain with quotient field K, and let L be a finite algebraic extension of K. (a) For any $v \in L$, show that there is a nonzero $a \in R$ such that av is integral over R. (b) Show that there is a basis $v_1, ..., v_n$ for L over K (as a vector space) such that each v_i is integral over R.

Solution

(a) Since L is algebraic over K, there are elements $a_i, b_i \in R$ such that

$$\frac{a_n}{b_n}v^n + ... + \frac{a_1}{b_1}v + \frac{a_0}{b_0} = 0.$$

Rewriting all of the fractions such that the have the same denominator, we get

$$\frac{a'_n}{b}v^n + ... + \frac{a'_1}{b}v + \frac{a'_0}{b} = 0$$

which implies that

$$a'_n v^n + ... + a'_1 v + a'_0 = 0.$$

Finally, multiplying both sides by $a_n'^{n-1}$ gives us

$$(a'_n v)^n + \cdots + (a'_1 a_n'^{n-2})(a'_n v) + a'_0 a_n'^{n-1} = 0$$

which shows that $a'_n v$ is integral over R. Finally, a'_n is nonzero because a_n is nonzero and none of the b_i are zero.

(b) Since L is a finite extension of K, there is a basis $u_1, ..., u_n$ for L over K. By part (a), there exist nonzero elements $a_i \in R$ such that $v_i := a_i u_i$ is integral over R. Since scaling the elements in a basis don't change the fact that the elements form a basis, then $v_1, ..., v_n$ is a basis for L over K composed only of elements which are integral over K.

Chapter 2

Affine Varieties

2.1 Coordinate Rings

2.1. Show that the map that associates to each $F \in k[X_1, ..., X_n]$ a polynomial function in $\mathcal{F}(V, k)$ is a ring homomorphism whose kernel is $I(V)$.

Solution Let $\varphi : k[X_1, \ldots, X_n] \to \mathcal{F}(V, k)$ be the map that sends each polynomial to its associated function on V. Let's show that it is a ring homomorphism. It is clear that $F(\lambda) = \lambda$ for all $\lambda \in k$. Moreover, since φ is simply a restriction map, then it preserves addition and multiplication of polynomials. Thus, it is a ring homomorphism.

Next, suppose that $F \in \ker \varphi$, i.e $\varphi(F) = 0$, then it follows that $F(x) = 0$ for all $x \in V$, and hence, $F \in I(V)$. Conversely if $F \in I(V)$, then $F(x) = 0$ for all $x \in V$ so $\varphi(F) = 0$ and hence, $F \in \ker \varphi$. Therefore, $\ker \varphi = I(V)$.

2.2. Let $V \subset \mathbb{A}^n$ be a variety. A *subvariety* of V is a variety $W \subset \mathbb{A}^n$ that is contained in V. Show that there is a natural one-to-one correspondence between algebraic subsets (resp. subvarieties, resp. points) of V and radical ideals (resp. prime ideals, resp. maximal ideals) of $\Gamma(V)$. (See Problems 1.22, 1.38.)

Solution This problem is strictly the same as Problem 1.38.

2.3. Let W be a subvariety of a variety V, and let $I_V(W)$ be the ideal of $\Gamma(V)$ corresponding to W. (a) Show that every polynomial function on V restricts to a polynomial function on W. (b) Show that the map from $\Gamma(V)$ to $\Gamma(W)$ defined in part (a) is a surjective homomorphism with kernel $I_W(V)$, so that $\Gamma(W)$ is isomorphic to $\Gamma(V)/I_V(W)$.

Solution

(a) Let $f \in \mathcal{F}(V, k)$ be a polynomial function, then there is a polynomial $F \in k[X_1, ..., X_n]$ such that $F(x) = f(x)$ for all $x \in V$. Hence, the function $f|_W \in \mathcal{F}(W, k)$ is also a polynomial function since $f|_W(x) = f(x) = F(x)$ for all $x \in W$.

(b) Let $\varphi : \Gamma(V) \to \Gamma(W)$ be the map that sends a polynomial function on V to its restriction on W. This is a ring homomorphism because, clearly $\varphi(\lambda) = \lambda$

for all $\lambda \in k$, and because it is a restriction map so it preserves addition and multiplication.

It is a surjective homomorphism because given any function $f \in \Gamma(W)$, we know that there exists a polynomial $F \in k[X_1, ..., X_n]$ such that $f(x) = F(x)$ for all $x \in W$. Hence, if we define $g \in \Gamma(V)$ by $g(x) = F(x)$ for all $x \in V$, then clearly $\varphi(g) = f$. It follows that φ is surjective.

If $f \in \ker(\varphi)$, then it is equivalent to say that f is sent to 0 by φ. Equivalently, it means that $f(x) = 0$ for all $x \in W$. But by definition, this is equivalent to $f \in I_V(W)$ since $I_V(W)$ is the projection of $I(V)$ in $\Gamma(V)$. Thus, $\ker(\varphi) = I_V(W)$. Therefore, by the First Isomorphism Theorem: $\Gamma(V)/I_V(W) = \Gamma(W)$.

2.4. Let $V \subset \mathbb{A}^n$ be a nonempty variety. Show that the following are equivalent: (i) V is a point; (ii) $\Gamma(V) = k$; (iii) $\dim_k \Gamma(V) < \infty$.

Solution If V is a point $(a_1, ..., a_n)$, then $I(V) = (X_1 - a_1, ..., X_n - a_n)$ which implies that $\Gamma(V) = k[X_1, ..., X_n]/(X_1 - a_1, ..., X_n - a_n) = k$. Hence, (i) implies (ii).

Next, if $\Gamma(V) = k$, then clearly $\dim_k \Gamma(V) = \dim_k k = 1 < \infty$. Hence, (ii) implies (iii).

Finally, if $\dim_k \Gamma(V) < \infty$, then $V(I(V)) = V$ is a finite set. But since V is irreducible, then it must be a point. Thus, (iii) implies (i). Therefore, the three propositions are equivalent.

2.5. Let F be an irreducible polynomial in $k[X, Y]$, and suppose F is monic in Y: $F = Y^n + a_1(X)Y^{n-1} + \cdots + a_n(X)$, with $n > 0$. Let $V = V(F) \subset \mathbb{A}^2$. Show that the natural homomorphism from $k[X]$ to $\Gamma(V) = k[X, Y]/(F)$ is one-to-one, so that $k[X]$ may be regarded as a subring of $\Gamma(V)$; show that the residues $\overline{1}, \overline{Y}, ..., \overline{Y}^{n-1}$ generate $\Gamma(V)$ over $k[X]$ as a module.

Solution Both propositions follow from the fact that $\Gamma(V) = k[X, Y]/(F) = k[X][Y]/(F) \cong k[X][Y]_{<n}$ which denotes the ring of polynomials in Y of degree at most $n - 1$ (addition and multiplication modulo F) with coefficients in $k[X]$. Hence, $k[X]$ is simply the subring of "constant" polynomials; and the elements $\overline{1}, \overline{Y}, ..., \overline{Y}^{n-1}$ generate $\Gamma(V)$ because the elements in $\Gamma(V)$ are polynomials of Y of degree at most $n - 1$.

2.2 Polynomial Maps

2.6. Let $\varphi : V \to W$, $\psi : W \to Z$. Show that $\widetilde{\psi \circ \varphi} = \tilde{\phi} \circ \tilde{\psi}$. Show that the composition of polynomial maps is a polynomial map.

Solution For all functions $f \in \mathcal{F}(Z, k)$, we have

$$(\widetilde{\psi \circ \varphi})(f) = f \circ \psi \circ \varphi = \tilde{\varphi}(f \circ \psi) = \tilde{\varphi}(\tilde{\psi}(f)) = (\tilde{\phi} \circ \tilde{\psi})(f).$$

Thus, $\widetilde{\psi \circ \varphi} = \tilde{\phi} \circ \tilde{\psi}$.

Next, suppose that $V \subset \mathbb{A}^n$, $W \subset \mathbb{A}^m$ and $Z \subset \mathbb{A}^k$. If φ and ψ are polynomial maps, then there exist polynomials $T_1, ..., T_m \in k[X_1, ..., X_m]$ and polynomials $F_1, ...,$ $F_k \in k[X_1, ..., X_m]$ such that $\varphi(x) = (T_1(x), ..., T_m(x))$ for all $x \in \mathbb{A}^n$ and $\psi(x) = (F_1(x), ..., F_k(x))$ for all $x \in \mathbb{A}^m$. It follows that

$$(\psi \circ \varphi)(x) = \psi(T_1(x), ..., T_m(x)) = (F_1(T_1(x), ..., T_m(x)), ..., F_k(T_1(x), ..., T_m(x)))$$

for all $x \in \mathbb{A}^n$. For all $i = 1, ..., k$, the map that sends $x \in \mathbb{A}^n$ to $F_i(T_1(x), ..., T_m(x))$ is a polynomial because the composition of polynomials is also a polynomial. Therefore, the composition of two polynomial maps is a polynomial map.

2.7. If $\varphi : V \to W$ is a polynomial map, and X is an algebraic subset of W, show that $\varphi^{-1}(X)$ is an algebraic subset of V. If $\varphi^{-1}(X)$ is irreducible, and X is contained in the image of φ, show that X is irreducible. This gives a useful test for irreducibility.

Solution Let $X = V(F_1, ..., F_r)$, then $x \in \varphi^{-1}(X)$ iff $\varphi(x) \in X$, iff $F_i(\varphi(x)) = 0$ for all $i = 1, ..., r$. Therefore, $\varphi^{-1}(X) = V(F_1 \circ \varphi, ..., F_r \circ \varphi)$ which shows that $\varphi^{-1}(X)$ is an algebraic set ($F_i \circ \varphi$ is a polynomial for all i). Suppose now that $\varphi^{-1}(X)$ is irreducible, and X is contained in the image of φ. If $X = X_1 \cup X_2$ where X_1, X_2 are algebraic sets such that X_i is not contained in X_j for $i \neq j$, then $\varphi^{-1}(X) = \varphi^{-1}(X_1) \cup \varphi^{-1}(X_2)$. By the first part of this exercise, we have that both $\varphi^{-1}(X_1)$ and $\varphi^{-1}(X_2)$ are algebraic sets. Moreover, the fact that X is contained in the image of φ implies that $\varphi^{-1}(X_i) \subset \varphi^{-1}(X_j)$ for $i \neq j$ can never happen. Thus, $\varphi^{-1}(X)$ is not irreducible, a contradiction. Therefore, by contradiction, X is irreducible.

2.8. (a) Show that $\{(t, t^2, t^3) \in \mathbb{A}^3(k) \mid t \in k\}$ is an affine variety. (b) Show that $V(XZ - Y^2, YZ - X^3, Z^2 - X^2Y) \subset \mathbb{A}^3(\mathbb{C})$ is a variety. (*Hint:* $Y^3 - X^4, Z^3 - X^5, Z^4 - Y^5 \in I(V)$. Find a polynomial map from $\mathbb{A}^1(\mathbb{C})$ onto V.)

Solution

(a) Problem 1.33.

(b) Define the polynomial map $\varphi : \mathbb{A}^1(\mathbb{C}) \to \mathbb{A}^3(\mathbb{C})$ that maps a complex number t to (t^3, t^4, t^5). It is easy to verify that $\varphi(t) \in V(I)$ for all $t \in \mathbb{C}$ (which implies that $\varphi^{-1}(V(I)) = \mathbb{A}^1(\mathbb{C})$). Let's show that $V(I)$ is in the image of φ. If $(x, y, z) \in V(I)$, then $xz = y^2$, $yz = x^3$, and $z^2 = x^2y$. If $x = 0$ or $y = 0$, then it is clear that $(x, y, z) = (0, 0, 0) \in \operatorname{im} \varphi$. Otherwise, we have that

$z = y^2/x$ and $z = x^3/y$ which implies that $y^2/x = x^3/y$, and hence, $y^3 = x^4$. If we let t be a cube root of x, then $y^3 = t^{12}$. Since t can be any cube root of x, then we can choose, in particular the cube root such that $y = t^4$. Finally, since $yz = x^3$, then $z = t^9/t^4 = t^5$. Thus, $(x, y, z) = \varphi(t)$ which implies that $V(I) \subset \operatorname{im}\varphi$.

Next, since $\varphi^{-1}(V(I)) = \mathbb{A}^1(\mathbb{C})$ and $\mathbb{A}^1(\mathbb{C})$ is irreducible, then by Problem 2.7, $V(I)$ is irreducible.

2.9. Let $\varphi : V \to W$ be a polynomial map of affine varieties, $V' \subset V$, $W' \subset W$ subvarieties. Suppose $\varphi(V') \subset W'$. (a) Show that $\tilde{\varphi}(I_W(W')) \subset I_V(V')$ (see Problem 2.3). (b) Show that the restriction of φ gives a polynomial map from V' to W'.

Solution

(a) Let $\tilde{\varphi}(f) = f \circ \varphi \in \tilde{\varphi}(I_W(W'))$ with $f \in I_W(W')$, then for all $x \in V'$, $\varphi(x) \in W'$ which implies that $(f \circ \varphi)(x) = 0$ since $f \in I_W(W')$. It follows that $\tilde{\varphi}(f) \in I_V(V')$, and hence, $\tilde{\varphi}(I_W(W')) \subset I_V(V')$.

(b) Follows directly from the fact that $\varphi(V') \subset W'$, and the fact that φ is a polynomial map.

2.10. Show that the *projection map* $\mathrm{pr} = \mathbb{A}^n \to \mathbb{A}^r$, $n \geq r$, defined by $\mathrm{pr}(a_1, ..., a_n) = (a_1, ..., a_r)$ is a polynomial map.

Solution If we let $P_i \in k[X_1, ..., X_n]$ be the polynomial $P_i(X_1, ..., X_n) = X_i$ for all $i = 1, ..., r$, then clearly $\mathrm{pr} = (P_1, ..., P_r)$.

2.11. Let $f \in \Gamma(V)$, V a variety $\subset \mathbb{A}^n$. Define

$$G(f) = \{(a_1, ..., a_n, a_{n+1}) \in \mathbb{A}^{n+1} \mid (a_1, ..., a_n) \in V \text{ and } a_{n+1} = f(a_1, ..., a_n)\},$$

the *graph* of f. Show that $G(f)$ is an affine variety, and that the map $(a_1, ..., a_n) \mapsto (a_1, ..., a_n, f(a_1, ..., a_n))$ defines an isomorphism of V with $G(f)$. (Projection gives the inverse.)

Solution If $V = V(F_1, ..., F_r)$, then $G(f) = V(F_1, ..., F_r, X_{n+1} - f(X_1, ..., X_n))$; hence, it is an algebraic set. Next, consider the map $\varphi : V \to \mathbb{A}^{n+1}$ that maps $(a_1, ..., a_n)$ to $(a_1, ..., a_n, f(a_1, ..., a_n))$. This is a polynomial map with image equal to $G(f)$. Since $\varphi^{-1}(G(f)) = V$ is irreducible, then it follows that $G(f)$ is irreducible by Problem 2.7.

Next, let's show that φ is an isomorphism. Let $\psi : G(f) \to V$ be the projection map. This is a polynomial map by Problem 2.10. Let $v \in V$, then $\psi(\varphi(v)) = \psi(v, f(v)) = v$; let $(x, f(x)) \in G(f)$, then $\varphi(\psi(x, f(x))) = \varphi(x) = (x, f(x))$. Therefore, φ is an isomorphism.

2.12. (a) Let $\varphi : \mathbb{A}^1 \to V = V(Y^2 - X^3) \subset \mathbb{A}^2$ be defined by $\varphi(t) = (t^2, t^3)$. Show that although φ is a one-to-one, onto polynomial map, φ is not an isomorphism.

(*Hint:* $\tilde{\varphi}(\Gamma(V)) = k[T^2, T^3] \subset k[T] = \Gamma(\mathbb{A}^1)$.) (b) Let $\varphi : \mathbb{A}^1 \to V = V(Y^2 - X^2(X+1))$ be defined by $\varphi(t) = (t^2 - 1, t(t^2 - 1))$. Show that φ is one-to-one and onto, except that $\varphi(\pm 1) = (0, 0)$.

Solution

(a) Clearly, φ is a polynomial map. If $\varphi(s) = \varphi(t)$, then $s^2 = t^2$ and $s^3 = t^3$. If $t = 0$, then $s = 0$ and so $t = s$. Otherwise, since $s^2 = t^2$, then $s = \pm t$. If $s = -t$, then cubing both sides gives us $s^3 = -t^3$ which implies that $t^3 = -t^3$, and hence, $t = 0$, a contradiction. It follows that $s = -t$ cannot hold, and hence, $s = t$. Therefore, φ is one-to-one. To show that φ is onto, let $(x, y) \in V$, then $y^2 = x^3$. If we let t be a square root of x, then we get that $y^2 = t^6$. It follows that $y = \pm t^3 = (\pm t)^3$. If $y = t^3$, then we are done since $(x, y) = \varphi(t)$; if $y = (-t)^3$, then we get that $(x, y) = \varphi(-t)$. It follows that φ is onto.

Suppose that there is a polynomial map $\psi : V \to \mathbb{A}^1$ such that $\psi \circ \varphi = id$, then $\tilde{\varphi} \circ \tilde{\psi} = \tilde{id} = id : k[X] \to k[X]$ by Problem 2.6 using the fact that $\Gamma(\mathbb{A}^1) = k[X]$. But this implies that $\operatorname{im} \tilde{\varphi} = k[X]$ which is impossible since $\operatorname{im} \tilde{\varphi} = k[X^2, X^3] \not\ni X$. Therefore, ψ cannot exist, and φ cannot be an isomorphism.

(b) If $\varphi(s) = \varphi(t)$, then $s^2 - 1 = t^2 - 1$ and $s(s^2 - 1) = t(t^2 - 1)$. Suppose that $s^2 - 1 = t^2 - 1 \neq 0$, then $s(s^2 - 1) = t(t^2 - 1)$ directly implies that $s = t$. The case $s^2 - 1 = t^2 - 1 = 0$ implies that $s = \pm 1$ and $t = \pm 1$ which we don't consider here. Next, let $(x, y) \in V$, then $y^2 = x^2(x + 1)$. If we let t be a root of the equation $t^2 - 1 = x$, then $y^2 = (t^2 - 1)^2(t^2 - 1 + 1) = [t(t^2 - 1)]^2$. It follows that $y = \pm t(t^2 - 1)$. Equivalently, $(x, y) = \varphi(\pm t)$ which, in both cases, implies that $(x, y) \operatorname{im} \varphi$. Therefore, φ is one-to-one and onto, except at ± 1.

2.13. Let $V = V(X^2 - Y^3, Y^2 - Z^3) \subset \mathbb{A}^3$ as in Problem 1.40, $\overline{\alpha} : \Gamma(V) \to k[T]$ induced by the homomorphism α of that problem. (a) What is the polynomial map f from $\mathbb{A}^1$ to V such that $\tilde{f} = \overline{\alpha}$? (b) Show that f is one-to-one and onto, but not an isomorphism.

Solution First, recall that $\alpha : k[X, Y, Z] \to k[T]$ is defined by $\alpha(X) = T^9$, $\alpha(Y) = T^6$, and $\alpha(Z) = T^4$. Since $\ker \alpha = I(V)$, then we have an induced homomorphism $\overline{\alpha} : \Gamma(V) \to k[T]$ such that $\overline{\alpha} \circ \pi = \alpha$ where $\pi : k[X, Y, Z] \to \Gamma(V)$ is the projection map.

(a) By following the proof of Proposition 1 (or simply by looking at the definition of α), we get that $f : \mathbb{A}^1 \to V$ defined by $f(T) = (T^9, T^6, T^4)$ is the polynomial map that satisfies $\tilde{f} = \varphi$.

(b) If $f(s) = f(t)$, then $s^k = t^k$ for $k = 4, 6, 9$. We can assume that $s, t \neq 0$ since otherwise, $s = t$. From $s^4 = t^4$, we get that $s = i^m t$ for some integer $m = 0, 1, 2, 3$ where i denotes a root of -1. Taking both sides to the sixth power gives us $s^6 = (-1)^m t^6$. But since $s^6 = t^6$, then $t^6 = (-1)^m t^6$ which implies that $(-1)^m = 1$, and hence, that $m = 0, 2$. It follows that $s = (-1)^p t$ for some integer $p = 0, 1$. Next, taking the previous equation to the power of nine gives us that $s^9 = (-1)^p t^9$; but since $s^9 = t^9$, then $t^9 = (-1)^p t^9$, and hence $(-1)^p = 1$. It follows that $s = t$. Thus, f is one-to-one.

Let $(x, y, z) \in V$, then $x^2 = y^3$ and $y^2 = z^3$. If we let t be a fourth root of z, then $y^2 = t^{12}$ which implies that $y = \pm t^6$, and hence, that $y = (i^k t)^6$ where i is a root of $\sqrt{-1}$, and k is any of $0, 2$ or $1, 3$. Similarly, if we do the same with the ninth power, we get that $x = (i^k t)^9$ for some k which will correspond to the case of the sixth power. It follows that $(x, y, z) = f(i^k t)$, and hence, f is onto V. Finally, let's show that f is not an isomorphism. Suppose that there is a polynomial map $g : V \to \mathbb{A}^1$ such that $g \circ f = id$, then $\tilde{f} \circ \tilde{g} = \tilde{id} = id : k[T] \to k[T]$ by Problem 2.6 using the fact that $\Gamma(\mathbb{A}^1) = k[T]$. But this implies that $\operatorname{im} \tilde{f} = k[T]$ which is impossible since $\operatorname{im} \tilde{f} = k[T^9, T^6, T^4] \not\ni T$. Therefore, g cannot exist, and hence, f cannot be an isomorphism.

2.3 Coordinate Changes

2.14. A set $V \subset \mathbb{A}^n(k)$ is called a *linear subvariety* of $\mathbb{A}^n(k)$ if $V = V(F_1, ..., F_r)$ for some polynomials F_i of degree 1. (a) Show that if T is an affine change of coordinates on $\mathbb{A}^n$, then V^T is also a linear subvariety of $\mathbb{A}^n(k)$. (b) If $V \neq \varnothing$, show that there is an affine change of coordinates T of $\mathbb{A}^n$ such that $V^T = V(X_{m+1}, ..., X_n)$. (*Hint:* use induction on r.) So V is a variety. (c) Show that the m that appears in part (b) is independent of the choice of T. It is called the dimension of V. Then V is then isomorphic (as a variety) to $\mathbb{A}^m(k)$. (*Hint:* Suppose there were an affine change of coordinates T such that $V(X_{m+1}, ..., X_n)^T = V(X_{s+1}, ..., X_n)$, $m < s$; show that $T_{m+1}, ..., T_n$ would be dependent.)

Solution

(a) If we denote $I = (F_1, ..., F_r)$, then I^T is generated by the elements F^T with $F \in I$. But notice that if $F = \sum_i a_i F_i$, then

$$F^T = \tilde{T}(\sum_i a_i F_i) = \sum_i \tilde{T}(a_i)\tilde{T}(F_i) = \sum_i a_i^T F_i^T$$

which implies that $I^T = (F_1^T, ..., F_r^T)$. Since the F_i's and the T_i's are all linear, then the F_i^T are linear, and hence, $V^T = V(F_1^T, ..., F_r^T)$ is a linear subvariety.

(b) Let's prove it by induction on $r \geq 1$. When $r = 1$, we have that $V = V(f)$ where $\deg(f) = 1$. If we write $f = a_0 + \sum_i a_i X_i$, then we must have at least one i such that $a_i \neq 0$, call it $i = i_0$. Define the polynomial map $T = (T_1, ..., T_n)$ where $T_i = 0$ for all $i \neq i_0$, and $T_{i_0} = a_{i_0}^{-1}(X_n - a_0)$, then

$$V^T = V(f^T) = V\left(a_0 + \sum_i a_i T_i\right) = V\left(a_0 + a_{i_0}[a_{i_0}^{-1}(X_n - a_0)]\right) = V(X_n).$$

Hence, the base case holds. Next, suppose that the statement holds for some $r = k$ and consider the case where $V = V(f_1, \ldots, f_k, f_{k+1})$. By the induction hypothesis, there is a polynomial T such that

$$V^T = V(f_1^T, \ldots, f_k^T) \cap V(f_{k+1}^T) = V(X_{m+1}, ..., X_n, f_{k+1}^T).$$

Now, notice that if we write $f = a_0 + \sum_{i=1}^n a_i X_i$, then $(X_{m+1}, ..., X_n, f_{k+1}^T) = (X_{m+1}, ..., X_n, g)$ where $g = a_0 + \sum_{i=1}^m a_i X_i$. Hence: $V^T = V(g, X_{m+1}, ..., X_n)$. If $g = 0$, we are done, if $g \neq 0$, then there is a $i = i_0 \leq m$ such that $a_{i_0} \neq 0$. Thus, if we define the polynomial map $U = (U_1, ..., U_n)$ by $U_i = 0$ for all $1 \leq i \neq i_0 \leq m$, $U_{i_0} = a_{i_0}^{-1}(X_m - a_0)$, and $U_i = X_i$ for all $m+1 \leq i \leq n$. Thus, by construction:

$$V^{T \circ U} = V(g \circ U, X_{m+1} \circ U, ..., X_n \circ U) = V(X_m, X_{m+1}, ..., X_n).$$

Thus, the case $r = k + 1$ also holds. Therefore, by induction, the proposition is proved for all $r \geq 1$.

(c) Suppose there exist affine change of coordinates U_1 and U_2 such that

$$V^{U_1} = V(X_{m+1}, \ldots, X_n) \qquad \text{and} \qquad V^{U_2} = V(X_{s+1}, \ldots, X_n)$$

with $m < s$, then equivalently, there is a change of coordinates T such that $V(X_{m+1}, ..., X_n)^T = V(X_{s+1}, ..., X_n)$. If we write $T = (T_1, ..., T_n)$, we get that $V(T_{m+1}, ..., T_n) = V(X_{s+1}, ..., X_n)$. By the Nullstellensatz, and the fact that $(X_{s+1}, ..., X_n)$ is a prime ideal:

$$\{T_{m+1}, ..., T_n\} \subseteq \mathrm{Rad}(T_{m+1}, ..., T_n) = I(V(T_{m+1}, ..., T_n)) = (X_{s+1}, ..., X_n).$$

Thus, for all $m+1 \leq i \leq n$, $T_i \in (X_{s+1}, ..., X_n)$. Equivalently,

$$T_i = \alpha_{s+1,i} X_{s+1} + \cdots + \alpha_{n,i} X_n$$

for some scalars $\alpha_{j,i} \in k$, and for all $m+1 \leq i \leq n$ (technically, the coefficients should be polynomials but we can deduce that there must be such scalars by the fact that T_i has degree 1). Now, recall that T is an affine change of coordinates so its linear part must be invertible. Equivalently, the matrix form of its linear part must be invertible, and hence, its rows must be linearly independent vectors. But the above equation shows that the last $n-m$ rows of that matrix are of the form $(0, ..., 0, \alpha_{s+1,i}, ..., \alpha_{n,i})$. In other words, the vectors

$$\begin{pmatrix} 0 \\ \vdots \\ 0 \\ \alpha_{s+1,m+1} \\ \vdots \\ \alpha_{n,m+1} \end{pmatrix}, \quad \begin{pmatrix} 0 \\ \vdots \\ 0 \\ \alpha_{s+1,m+2} \\ \vdots \\ \alpha_{n,m+2} \end{pmatrix}, \quad \ldots, \quad \begin{pmatrix} 0 \\ \vdots \\ 0 \\ \alpha_{s+1,n} \\ \vdots \\ \alpha_{n,n} \end{pmatrix}$$

are linearly independent. This is equivalent to the fact that the vectors

$$\begin{pmatrix} \alpha_{s+1,m+1} \\ \vdots \\ \alpha_{n,m+1} \end{pmatrix}, \quad \begin{pmatrix} \alpha_{s+1,m+2} \\ \vdots \\ \alpha_{n,m+2} \end{pmatrix}, \quad \ldots, \quad \begin{pmatrix} \alpha_{s+1,n} \\ \vdots \\ \alpha_{n,n} \end{pmatrix}$$

are linearly. But this is impossible since we have $n-m$ vectors in k^{n-s}, and $n-m > n-s$. Therefore, we found a contradiction that proves that m is independent of the choice of T.

We now need to prove that V is isomorphic to $\mathbb{A}^m(k)$ as a variety. Let T be an affine coordinates change such that $V^T = V_m := V(X_{m+1}, ..., X_n)$, then $T(V_m) = V$. It follows that T, restricted to V_m, is a bijection with image equal to V. Since the inverse of T is also an affine change of coordinates, then it follows that V and V_m are isomorphic as variety. Thus, we now need to show that V_m is isomorphic to $\mathbb{A}^m(k)$ as variety. Consider the polynomial map $U : \mathbb{A}^m(k) \to V_m$ defined by

$$U(X_1, ..., X_m) = (X_1, ..., X_m, 0, ..., 0).$$

Its inverse is the polynomial map $\pi : V_m \to \mathbb{A}^m(k)$ defined by

$$\pi(X_1, ..., X_m, X_{m+1}, ..., X_n) = (X_1, ..., X_m).$$

By construction, they are inverses of each other so U is an isomorphism of variety. Thus, V is isomorphic to V_m, which is in turn isomorphic to $\mathbb{A}^m(k)$ as a variety.

2.15. Let $P = (a_1, ..., a_n)$, $Q = (b_1, ..., b_n)$ be distinct points in $\mathbb{A}^n$. The *line* through P and Q is defined to be $\{a_1 + t(b_1 - a_1), ..., a_n + t(b_n - a_n) | t \in k\}$. (a) Show that if L is the line through P and Q, and T is an affine change of coordinates, then $T(L)$ is the line through $T(P)$ and $T(Q)$. (b) Show that a line is a linear subvariety of dimension 1, and that a linear subvariety of dimension 1 is the line through any of its two points. (c) Show that, in $\mathbb{A}^2$, a line is the same thing as a hyperplane. (d) Let $P, P' \in \mathbb{A}^2$, L_1, L_2 two distinct lines through P, L_1', L_2' are distinct lines through P'. Show that there is an affine change of coordinates T of $\mathbb{A}^2$ such that $T(P) = P'$ and $T(L_i) = L_i'$, $i = 1, 2$.

Solution

(a) First, rewrite the line through P and Q as $L = \{P + t(Q - P) | t \in k\}$. If we let $T = T' + \alpha$ be an affine change of coordinates where T' is the linear part, and $\alpha \in k^n$ is a constant, then

$$\begin{aligned} T(L) &= \{T(P + t(Q - P)) | t \in k\} \\ &= \{T'(P + t(Q - P)) + \alpha | t \in k\} \\ &= \{T'(P) + t(T'(Q) - T'(P)) + \alpha | t \in k\} \\ &= \{[T'(P) + \alpha] + t([T'(Q) + \alpha] - [T'(P) + \alpha]) | t \in k\} \\ &= \{T(P) + t(T(Q) - T(P)) | t \in k\} \end{aligned}$$

which is precisely the definition of the line through $T(P)$ and $T(Q)$.

(b) First, consider the affine change of coordinates T_1 that maps x to $x - P$ (a translation), then the image of L is the line L_1 through 0 with director vector $v_1 = Q - P$. Next, since v_1 is a non-zero vector, then we can find vectors $v_2, ..., v_n$ such that $v_1, ..., v_n$ forms a basis of k^n. From that, define the linear transformation T_2 that maps v_i to e_i where $e_1, ..., e_n$ is the standard basis. By construction, T_2 is an invertible linear transformation, so it is an affine change of coordinates. The image of L_1 under T_2 is the line $L_2 = V(X_2, ..., X_n)$, which is a linear subvariety. Thus, by construction, we have that $L = T_1^{-1}(T_2^{-1}(V)) = V^{T_2 \circ T_1}$ is also a linear subvariety (by Problem 2.14 part (a)).

Conversely, let L be a linear subvariety of dimension 1, then there is an affine change of coordinates T, such that $L^T = V(X_2, ..., X_n)$, or equivalently, such that $L = T(V(X_2, ..., X_n))$. Let P and Q be two points in L, then there are two points $(p, 0, ..., 0), (q, 0, ..., 0) \in V(X_2, ..., X_n)$ that are the respective preimages of these two points. Since P and Q are distinct, and T is a bijection, then p and q are also distinct. Hence, we can easily see that $V(X_2, ..., X_n)$ can be described as the line L_0 through the vectors $(p, 0, ..., 0)$ and $(q, 0, ..., 0)$. Finally, using part (a), we get that $L = T(L_0)$ must be the line through the points P and Q. Since $P \neq Q$ were arbitrary points on L, then we can say that a linear subvariety fo dimension 1 is the line through any two of its points.

(c) Let L be a line in $\mathbb{A}^2$, then by the previous part, we know that there exists an affine change of coordinates T such that $L^T = V(X_2)$, or equivalently, such that $L = T(V(X_2))$. Since T is a bijection such that its inverse is also an affine change of coordinates, we can rewrite the previous equation as

$$L = T(V(X_2)) = (T^{-1})^{-1}(V(X^2)) = V(X_2)^{T^{-1}}.$$

Hence, if we write $T^{-1} = (T_1, T_2)$, we get that $L = V(X_2 \circ T^{-1}) = V(T_2)$. Therefore, L is a hyperplane (T_2 has degree 1).

Conversely, let $V(f)$ be a hyperplane, then f is a polynomial of degree 1. By the proof of the base case of the induction in the solution of part (b) in Problem 2.14, we have that there exists an affine change of coordinates T such that $V(f)^T = V(X_2)$. Since the latter is the line through $(0,0)$ and $(1,0)$, and $T(V(X_2)) = V(f)$, then $V(f)$ must be a line. Therefore, in $\mathbb{A}^2$, hyperplanes and lines form the same class of objects.

(d) First, define the affine change of coordinates T_1 that acts as a translation by mapping the point P to the origin. Thus, the image of the lines L_1 and L_2 are now intersecting at the origin. Let v_1 and v_2 be director vectors of the lines $T(L_1)$ and $T(L_2)$, since the lines are distinct, the vectors form a basis of $\mathbb{A}^2$. Similarly, repeat the same process for the lines L_1' and L_2': let T_2 be an affine change of coordinates that maps P' to the origin (a translation), and let v_3, v_4 be the director vectors of the lines $T_2(L_1')$ and $T_2(L_2')$ respectively. Again, v_3, V_4 forms another basis of $\mathbb{A}^2$. Finally, consider the linear transformation that maps v_1 to v_3 and v_2 to v_4, then it is invertible, and so it is an affine change of coordinates. By construction, $T(T_1(L_1)) = T_2(L_1')$ and $T(T_1(L_2)) = T_2(L_2')$. Therefore, the affine change of coordinates $U = T_2^{-1} \circ T \circ T_1$ has the property that $U(P) = P'$ and $U(L_i) = L_i'$ for $i = 1, 2$.

2.16. Let $k = \mathbb{C}$. Give $\mathbb{A}^n(\mathbb{C}) = \mathbb{C}^n$ the usual topology (obtained by identifying $\mathbb{C}$ with $\mathbb{R}^2$, and hence $\mathbb{C}^n$ with $\mathbb{R}^{2n}$). Recall that a topological space X is *path-connected* if for any $P, Q \in X$, there is a continuous mapping $\gamma : [0,1] \to X$ such that $\gamma(0) = P$ and $\gamma(1) = Q$. (a) Show that $\mathbb{C} \setminus S$ is path-connected for any finite set S. (b) Let V be an algebraic set in $\mathbb{A}^n(\mathbb{C})$. Show that $\mathbb{A}^n(\mathbb{C}) \setminus V$ is path-connected. (*Hint:* If $P, Q \in \mathbb{A}^n(\mathbb{C}) \setminus V$, let L be the line through P and Q. Then $L \cap V$ is finite, and L is isomorphic to $\mathbb{A}^1(\mathbb{C})$.)

Solution

(a) Let $S = \{s_1, ..., s_n\}$ and consider the line segment L between P and Q. If $L \cap S = \varnothing$, then it suffices to take $\gamma(t) = p + t(q - p)$. Otherwise, take any point $l \in L$ and consider the straightline L_0 through l that is perpendicular to L. For all points $z \in L_0$, we can define the line segments $L_1^{(z)}$ between P and z, $L_2^{(z)}$ between z and Q. For two distinct $z_1, z_2 \in L_0$, we have that

$$(L_1^{(z_1)} \cup L_2^{(z_1)}) \cap (L_1^{(z_2)} \cup L_2^{(z_2)}) = \{P, Q\}.$$

Thus, if we define $L_z = L_1^{(z)} \cup L_2^{(z)}$, then there must be a $z_0 \in L_0$ such that $L_z \cap S = \varnothing$ since there are infinitely many points on L_0 but only finitely many in S. From that, define

$$\gamma(t) = \begin{cases} p + 2t(z_0 - p) & 0 \le t < \frac{1}{2}, \\ z_0 + 2t(q - z_0) & \frac{1}{2} \le t \le 1. \end{cases}.$$

After easily verifying that γ is continuous, we are done.

(b) Let $P, Q \in \mathbb{A}^n(\mathbb{C}) \setminus V$ and $L = \{P + t(Q - P) | t \in \mathbb{C}\}$ be the line through P and Q. If we write $V = V(f_1, ..., f_r)$, then

$$L \cap V = (L \cap V(f_1)) \cap V(f_2, ..., f_r) \subseteq L \cap V(f_1).$$

If we look at $L \cap V(f_1)$ more closely, we get that it is the set of solutions of the polynomial f_1 such that the input is in the line L. If we define $g(t) = P + t(Q - P)$, we get that the elements are precisely the image under g of the solutions of the function $f_1(g(t))$. But notice now that $f_1 \circ g$ is only one variable so it has finitely many solutions. Thus, $L \cap V(f_1)$ is finite, and hence, so is $L \cap V$.

Next, since L is a linear subvariety of dimension 1, then it is isomorphic to $\mathbb{A}^1(\mathbb{C})$ as a variety (Problem 2.14 part (c)). Hence, there is a bijective polynomial map $T : \mathbb{C} \to L$. Let $S = T^{-1}(L \cap V)$, then S is finite. By part (a), there is a path $\gamma : [0, 1] \to \mathbb{C} \setminus S$ from $P' = T^{-1}(P)$ to $Q' = T^{-1}(Q)$. It follows that $T \circ \gamma : [0, 1] \to L \setminus V \subseteq \mathbb{A}^n(\mathbb{C}) \setminus V$ is path from P to Q. Therefore, $\mathbb{A}^n(\mathbb{C})$ is path-connected.

2.4 Rational Functions and Local Rings

2.17. Let $V = V(Y^2 - X^2(X+1)) \subset \mathbb{A}^2$, and $\overline{X}$, $\overline{Y}$ the residues of X, Y in $\Gamma(V)$; let $z = \overline{Y}/\overline{X} \in k(V)$. Find the pole sets of z and of z^2.

Solution First, notice that $z^2 = \overline{X+1}$ so its pole set is the empty set. Let's prove that $(0,0)$ is the only element in the pole set of z. Since $z = \overline{Y}/\overline{X}$ and $z = \overline{X(X+1)}/\overline{Y}$, then $(0,0)$ is the only possible element in the pole set. If the pole set is empty, then z must be a rational function, and hence, we have the equality $\overline{Y}/\overline{X} = \overline{p(X,Y)}$ which is equivalent to

$$Y - Xp(X,Y) = q(X,Y)(Y^2 - X^2(X+1))$$

in $k[X,Y]$. Taking $X = 0$ gives us the equation $Y = q(0,Y)Y^2$ which is impossible. Therefore, the pole set of z is composed of $(0,0)$ only.

2.18. Let $\mathcal{O}_P(V)$ be the local ring of a variety V at a point P. Show that there is a natural one-to-one correspondence between the prime ideals in $\mathcal{O}_P(V)$ and the subvarieties of V that pass through P. (*Hint:* If I is prime in $\mathcal{O}_P(V)$, $I \cap \Gamma(V)$ is prime in $\Gamma(V)$, and I is generated by $I \cap \Gamma(V)$; use Problem 2.2.)

Solution By Problem 2.2, we know that there is a natural one-to-one correspondence between the subvarieties of V and the prime ideals of $\Gamma(V)$. If I is a prime ideal of $\mathcal{O}_P(V)$, then $I \cap \Gamma(V)$ is a prime ideal of $\Gamma(V)$. Conversely, if I is a prime ideal of $\Gamma(V)$, then viewed as a subset of $\mathcal{O}_P(V)$, it generates a prime ideal I' of $\mathcal{O}_P(V)$ (same idea as in the proof of Proposition 3). This establishes a natural one-to-one correspondence between the prime ideals of $\Gamma(V)$ and the prime ideals of $\mathcal{O}_P(V)$. Therefore, there is a natural one-to-one correspondence between the prime ideals in $\mathcal{O}_P(V)$ and the subvarieties of V.

2.19. Let f be a rational function on a variety V. Let $U = \{P \in V \mid f \text{ is defined at } p\}$. Then f defines a function from U to k. Show that this function determines f uniquely. So a rational function may be considered as a type of function, but only on the complement of an algebraic subset of V, not on V itself.

Solution Let f and g be two rational functions in $k(V)$ that have the same induced function, then $h = f - g$ is a rational function in $k(V)$ such that its induced function is the zero function on its domain. Hence, it suffices to show that if $h \in k(V)$ induces the zero function on its domain U, then it is the zero rational function. To do so, write $h = \overline{a}/\overline{b}$ where $\overline{a}, \overline{b} \in \Gamma(V)$ and such that $b(P) \neq 0$ for some $P \in V$ (i.e, h is defined on at least one point). By our assumption on h, we have that $a(P)/b(P) = 0$ whenever h is defined at p, or equivalently, whenever $b(P) \neq 0$. But notice that $a(P)/b(P) = 0$ implies that $a(P) = 0$. Hence, $b(P) \neq 0$ implies that $a(P) = 0$. It follows that the polynomial ab satisfies $a(P)b(P) = 0$ for all $P \in V$. Thus, $ab \in I(V)$, and hence, $\overline{a}\overline{b} = 0$ in $\Gamma(V)$. But since V is a variety, $\Gamma(V)$ is an integral domain so we must have $\overline{a} = 0$ or $\overline{b} = 0$. Since we know that $\overline{b}$ is non-zero on at least one point of V, then we get that $\overline{a} = 0$. It follows that $h = 0$ as a rational function. Therefore, $f = g$.

2.20. In the example given in this section, show that it is impossible to write $f = a/b$, where $a, b \in \Gamma(V)$, and $b(P) \neq 0$ for every P where f is defined. Show that the pole set of f is exactly $\{(x, y, z, w) | y = 0 \text{ and } w = 0\}$.

Solution First, suppose by contradiction that we can write $f = \overline{a}/\overline{b}$ where

$$b(x_0, y_0, z_0, w_0) = 0 \implies y_0 = w_0 = 0,$$

then $b(x_0, y_0, z_0, w_0) \neq 0$ whenever $y_0 \neq 0$ or $w_0 \neq 0$. Hence, if $y_0 \neq 0$, the polynomial $b(x, y_0, z, w)$ has no roots, and hence, it is equal to a non-zero constant. Similarly, when $w_0 \neq 0$, the polynomial b is constant. Let $(x_i, y_i, z_i, w_i) \in \mathbb{A}^4 \setminus V(y, w)$ for $i = 1, 2$. Without loss of generality, we can assume that $y_1, w_2 \neq 0$. Thus, we get that

$$b(x_1, y_1, z_1, w_1) = b(x_2, y_1, z_2, w_2) = b(x_2, y_2, z_2, w_2) =: c \in \mathbb{A}^1 \setminus \{0\}$$

using our previous remarks. This proves that b is constant on the set $\mathbb{A}^4 \setminus V(y, w)$. Now, fix $(x_0, z_0, w_0) \in \mathbb{A}^4$ and consider the polynomial $b_0(y) = b(x_0, y, z_0, w_0)$. By what we proved, we know that this polynomial is constant on the set $\mathbb{A}^1 \setminus \{0\}$. Our goal is to show that $b_0(0) = c$, and hence, that b_0 is constant over its domain. If $b_0(0) \neq 0$, then b_0 has no roots and so it is a constant polynomial. It follows that $b_0 \equiv c$. If $b_0(0) = 0$, then $b_0(y) = y^m b_1(y)$ where m is the multiplicity of the root, and b_1 is a polynomial that doesn't vanish at $y = 0$. Since the roots of b_1 are necessarily roots of b_0, then b_1 has no roots. Thus, it is a non-zero constant polynomial, giving us $b_0(y) = Cy^m$ where $C \neq 0$. But notice that k is algebraically closed so there is certainly an mth root of 2 which we will call α. Thus, $b_0(1) = C$ and $b_0(\alpha) = 2C$, contradicting the fact that b_0 is constant on $\mathbb{A}^1 \setminus \{0\}$. Hence, the only possible case is that $b_0 \equiv c$. Since it holds for all $(x_0, z_0, w_0) \in \mathbb{A}^3$, we get that $b \equiv c$. Thus, $f = \frac{1}{c}\overline{a} \in \Gamma(V)$. Let's now prove that this is impossible. Notice that $f = \overline{x}/\overline{y}$ implies that

$$x = \frac{1}{c} y a(x, y, z, w) + h(x, y, z, w)(xw - yz).$$

Taking $y = w = 0$ gives us $x = 0$, a contradiction. Therefore, by contradiction, such a polynomial b cannot exist.

Next, let's prove that the pole set of f is exactly $V(y, w)$. From the two expressions of f: $f = \overline{x}/\overline{y} = \overline{z}/\overline{w}$, we get that the pole set must be a subset of $V(y, w)$. Conversely, let $p = (p_x, 0, p_z, 0) \in V(y, w)$ and suppose by contradiction that we can write $f = \overline{a}/\overline{b}$ where $b(p) \neq 0$. From the equation $\overline{x}/\overline{y} = \overline{a}/\overline{b}$, we get

$$xb(x, y, z, w) = ya(x, y, z, w) + h_1(x, y, z, w)(xw - yz).$$

Taking $y = w = 0$ gives us

$$xb(x, 0, z, 0) = 0.$$

But this is impossible because we know that $b(x, 0, z, 0)$ is a non-zero polynomial ($b(p_x, 0, p_z, 0) \neq 0$). Thus, by contradiction, p must be in the pole set of f. It follows that the pole set of f is $V(y, w)$.

2.21. Let $\varphi : V \to W$ be a polynomial map of affine varieties, $\tilde{\varphi} : \Gamma(W) \to \Gamma(V)$ the induced map on coordinate rings. Suppose $P \in V$, $\varphi(P) = Q$. Show that $\tilde{\varphi}$ extends uniquely to a ring homomorphism (also written $\tilde{\varphi}$) from $\mathcal{O}_Q(W)$ to $\mathcal{O}_P(V)$. (Note that $\tilde{\varphi}$ may not extend to all of $k(W)$.) Show that $\tilde{\varphi}(\mathfrak{m}_Q(W)) \subset \mathfrak{m}_P(V)$.

Solution Define $\psi : \mathcal{O}_Q(W) \to \mathcal{O}_P(V)$ by $\psi(a/b) = \tilde{\varphi}(a)/\tilde{\varphi}(b) = (a \circ \varphi)/(b \circ \varphi)$. First, let's show that its maps to $\mathcal{O}_P(V)$. Let $a/b \in O_Q(W)$ such that $b(Q) \neq 0$, then $(b \circ \varphi)(P) = b(Q) \neq 0$. It follows that $\tilde{\varphi}(a)/\tilde{\varphi}(b) \in O_P(V)$. Next, we need to show that the map is well-defined. Suppose that $a/b = c/d \in O_Q(W)$, then $ad = bc$ as elements of $\Gamma(W)$. Since $\tilde{\varphi}$ is a morphism on $\Gamma(W)$, we get that $\tilde{\varphi}(a)\tilde{\varphi}(d) = \tilde{\varphi}(b)\tilde{\varphi}(c)$, and hence, $\tilde{\varphi}(a)/\tilde{\varphi}(b) = \tilde{\varphi}(c)/\tilde{\varphi}(d)$. Therefore, ψ is a well-defined map from $\mathcal{O}_Q(W)$ to $\mathcal{O}_P(V)$.

Now, we need to show that ψ is a ring homomorphism:

$$\psi(1) = \psi(1/1) = \tilde{\varphi}(1)/\tilde{\varphi}(1) = 1/1 = 1,$$

and

$$\begin{aligned}
\psi((a/b) + (c/d)) &= \psi((ad + bc)/bd) \\
&= \tilde{\varphi}(ad + bc)/\tilde{\varphi}(bd) \\
&= (\tilde{\varphi}(a)\tilde{\varphi}(d) + \tilde{\varphi}(b)\tilde{\varphi}(c))/\tilde{\varphi}(b)\tilde{\varphi}(d) \\
&= (\tilde{\varphi}(a)/\tilde{\varphi}(b)) + (\tilde{\varphi}(c)/\tilde{\varphi}(d)) \\
&= \psi(a/b) + \psi(c/d),
\end{aligned}$$

and

$$\begin{aligned}
\psi((a/b) \cdot (c/d)) &= \psi((ac)/(bd)) \\
&= \tilde{\varphi}(ac)/\tilde{\varphi}(bd) \\
&= (\tilde{\varphi}(a)\tilde{\varphi}(c))/(\tilde{\varphi}(b)\tilde{\varphi}(d)) \\
&= (\tilde{\varphi}(a)/\tilde{\varphi}(b)) \cdot (\tilde{\varphi}(c)/\tilde{\varphi}(d)) \\
&= \psi(a/b)\psi(c/d).
\end{aligned}$$

Next, we need to show that it extends $\tilde{\varphi}$. For all $a \in \Gamma(W) \subseteq O_Q(W)$, we have

$$\psi(a) = \psi(a/1) = \tilde{\varphi}(a)/\tilde{\varphi}(1) = \tilde{\varphi}(a)/1 = \tilde{\varphi}(a).$$

Now, let $\psi_0 : \mathcal{O}_Q(W) \to \mathcal{O}_P(V)$ be a ring homomorphism that extends $\tilde{\varphi}$, then for all a/b in $\mathcal{O}_Q(W)$:

$$\begin{aligned}
&\tilde{\varphi}(b)\psi_0(a/b) = \psi_0(b)\psi_0(a/b) = \psi_0(b \cdot (a/b)) = \psi_0(a) = \tilde{\varphi}(a) \\
&\implies \psi_0(a/b) = \tilde{\varphi}(a)/\tilde{\varphi}(b) = \psi(a/b).
\end{aligned}$$

Therefore, $\psi_0 = \psi$, and hence, ψ is unique. Thus, we will now define $\tilde{\varphi}$ to be ψ.

Finally, let $a/b \in \mathfrak{m}_Q(W)$, then $a(Q) = 0$ which gives us

$$\tilde{\varphi}(a/b)(P) = (a \circ \varphi)(P)/(b \circ \varphi)(P) = a(Q)/b(Q) = 0$$

so $\tilde{\varphi}(a/b) \in \mathfrak{m}_P(V)$. Thus, $\tilde{\varphi}(\mathfrak{m}_Q(W)) \subset \mathfrak{m}_P(V)$.

2.22. Let $T : \mathbb{A}^n \to \mathbb{A}^n$ be an affine change of coordinates, $T(P) = Q$. Show that $\tilde{T} : \mathcal{O}_Q(\mathbb{A}^n) \to \mathcal{O}_P(\mathbb{A}^n)$ is an isomorphism. Show that $\tilde{T}$ induces an isomorphism from $\mathcal{O}_Q(V)$ to $\mathcal{O}_P(V^T)$ if $P \in V^T$, for V a subvariety of $\mathbb{A}^n$.

Solution Since T is an isomoprhism from $\mathbb{A}^n$ to $\mathbb{A}^n$, then $\tilde{T} : \Gamma(\mathbb{A}^n) \to \Gamma(\mathbb{A}^n)$ is also an isomorphism with inverse $\tilde{T}^{-1} : \Gamma(\mathbb{A}^n) \to \Gamma(\mathbb{A}^n)$. By Problem 2.21, both $\tilde{T}$ and $\tilde{T}$ must act by

$$\tilde{T}(a/b) = \tilde{T}(a)/\tilde{T}(b), \quad \text{and} \quad \tilde{T}^{-1}(c/d) = \tilde{T}^{-1}(c)/\tilde{T}^{-1}(d).$$

Hence, for all $a/b \in \mathcal{O}_Q(\mathbb{A}^n)$ and $c/d \in \mathcal{O}_P(\mathbb{A}^n)$:

$$\tilde{T}^{-1}(\tilde{T}(a/b)) = \tilde{T}^{-1}(\tilde{T}(a))/\tilde{T}^{-1}(\tilde{T}(b)) = a/b$$

and

$$\tilde{T}(\tilde{T}^{-1}(c/d)) = \tilde{T}(\tilde{T}^{-1}(c))/\tilde{T}(\tilde{T}^{-1}(d)) = c/d.$$

Therefore, $\tilde{T} : \mathcal{O}_Q(\mathbb{A}^n) \to \mathcal{O}_P(\mathbb{A}^n)$ is an isomorphism. The proof that $\tilde{T}$ induces an isomorphism from $\mathcal{O}_Q(V)$ to $\mathcal{O}_P(V^T)$ is the same as the one we just did since T is an isomorphism from V^T to V.

2.5 Discrete Valuation Rings

2.23. Show that the order function on K is independent of the choice of uniformizing parameter.

Solution Let t_1 and t_2 be two uniformizing parameters for R. Since t_2 is clearly non-zero, then we must have that $t_2 = ut_1^n$ for some unit u and $n \geq 1$. Similarly, we also have $t_1 = vt_2^m$ for some unit v and $m \geq 1$. It follows that $t_2 = uv^n t_2^{nm}$. By the uniqueness of the expressions of the form $u_0 t_2^{n_0}$, we get that $nm = 1$, and hence, that $n = 1$. It follows that $t_2 = ut_1$ for some unit u.

Next, let $z \in R \setminus \{0\}$, then we have that $z = u_1 t_1^{n_1}$ and $z = u_2 t_2^{n_2}$ for some units u_1, u_2 and integers $n_1, n_2 \geq 0$. Since $t_2 = ut_1$, then $u_1 t_1^{n_1} = u_2 t_2^{n_2}$ implies that

$$u_1 t_1^{n_1} = (u_2 u^{n_2}) t_1^{n_2}.$$

Since $u_2 u^{n_2}$ is unit, we get that $n_1 = n_2$ by uniqueness. Therefore, the order of z doesn't depend on the uniformizing parameter.

2.24. Let $V = \mathbb{A}^1$, $\Gamma(V) = k[X]$, $K = k(V) = k(X)$. (a) For each $a \in k = V$, show that $\mathcal{O}_a(V)$ is a DVR whith uniformizing parameter $t = X - a$. Show that $\mathcal{O}_\infty = \{F/G \in k(X) \mid \deg(G) \geq \deg(F)\}$ is also a DVR, with uniformizing parameter $t = 1/X$.

Solution Since $k[X]$ is a UFD, every element in $\mathcal{O}_a(V) \subset k(X)$ has a unique representation P/Q where P and Q have no common factors. By properties of polynomials of one variable, we can write $P(X) = (X - a)^m P_0(X)$ where $m \geq 0$ and such that $P_0(a) \neq 0$. Hence, $P/Q = (P_0/Q)(X - a)^m$. The uniqueness of this expression comes from the unique factorization in $k[X]$. The rational function P_0/Q is a unit since $Q/P_0 \in \mathcal{O}_a(V)$. Since the element $t = X - a$ is irreducible, then $\mathcal{O}_a(V)$ is a DVR with uniformizing parameter $t = X - a$.

By the unique factorization in $k[X]$, we can write each element F/G in $\mathcal{O}_\infty$ as $(F_0/G_0) \cdot (1/X)^m$ where F_0 and G_0 are non divisible by X, and where $m \in \mathbb{Z}$. Hence, $\deg(G_0) + m \geq \deg(F_0)$. Next, define $k = \deg(F_0) - \deg(G_0)$, then $(m - k) + \deg(G_0) + k \geq \deg(F_0)$ and $\deg(G_0) + k = \deg(F_0)$. From this, we get that we can write

$$\frac{F}{G} = \left(\frac{F_0}{G_0} \cdot X^{-k}\right) \cdot \left(\frac{1}{X}\right)^{m-k}$$

If $m - k < 0$, then

$$\deg(F_0) \leq \deg(G_0) + m < \deg(G_0) + k = \deg(F_0)$$

which is a contradiction. Thus, $m - k \geq 0$. Moreover, the term $(F_0/G_0)X^{-k}$ is a unit in $\mathcal{O}_\infty$ because the degree of the numerator is equal to the degree of the denominator. Next, suppose that we can write

$$\frac{1}{X} = \frac{P_1}{Q_1} \cdot \frac{P_2}{Q_2}$$

where $(P_1/Q_1), (P_2/Q_2) \in \mathcal{O}_\infty$ and none are units, then $\deg(P_1) < \deg(Q_1)$ and $\deg(P_2) < \deg(Q_2)$. Equivalently, $\deg(P_1)+1 \leq \deg(Q_1)$ and $\deg(P_2)+1 \leq \deg(Q_2)$. Hence:

$$2 + \deg(P_1) + \deg(P_2) \leq \deg(Q_1) + \deg(Q_2).$$

Now, if we rearrange the above equation, we obtain

$$Q_1Q_2 = XP_1P_2$$

from which we get

$$\begin{aligned}\deg(Q_1) + \deg(Q_2) &= 1 + \deg(P_1) + \deg(P_2)\\ &< 2 + \deg(P_1) + \deg(P_2)\\ &\leq \deg(Q_1) + \deg(Q_2),\end{aligned}$$

a contradiction. Hence, $1/X$ is irreducible in $\mathcal{O}_\infty$. Finally, suppose that

$$\frac{F_0}{G_0}\left(\frac{1}{X}\right)^{n_0} = \frac{F_1}{G_1}\left(\frac{1}{X}\right)^{n_1}$$

where $(F_0/G_0), (F_1/G_1) \in \mathcal{O}_\infty$ are units, then $\deg(F_0) = \deg(G_0)$ and $\deg(F_1) = \deg(G_1)$. If we rearrange the equation, we get

$$F_0G_1X^{n_1} = F_1G_0X^{n_0}$$

which gives us

$$\deg(F_0) + \deg(G_1) + n_1 = \deg(F_1) + \deg(G_0) + n_0.$$

Hence, $n_0 = n_1$, and $F_0/G_0 = F_1/G_1$. Therefore, the expressions of that form are unique, and so $\mathcal{O}_\infty$ is a DVR with uniformizing paramter $t = 1/X$.

2.25. Let $p \in \mathbb{Z}$ be a prime number. Show that $\{r \in \mathbb{Q} \mid r = a/b,\ a, b \in \mathbb{Z},\ p$ doesn't divide $b\}$ is a DVR with quotient field $\mathbb{Q}$.

Solution Let's first show that set we are interested in, which we will call R, is a DVR. First, R is a ring because it is a subset of the ring $\mathbb{Q}$ that contains the additive and multiplicative identities, and that is closed under addition and multiplication. Moreover, it is a domain since it is a subring of a domain. Let $a/b \in R$, then we can write $a = a_0p^n$ where $n \geq 0$ and p doesn't divide a_0. Hence, $a/b = (a_0/b)p^n$ and a_0/b is a unit since $b/a_0 \in R$.

Next, let $a_i/b_i \in R^\times$ and $n_i \geq 0$ for $i = 0, 1$, then none of a_0, a_1, b_0, b_1 are divisible by p. Suppose that

$$\frac{a_0}{b_0}p^{n_0} = \frac{a_1}{b_1}p^{n_1},$$

then

$$a_0b_1p^{n_0} = a_1b_0p^{n_1}.$$

By the Fundamental Theorem of Arithmetic, $n_0 = n_1$, and hence, $a_0/b_0 = a_1/b_1$. Therefore, the expressions of that form are unique. Finally, the element p is irreducible in R because if $p = (a_0/b_0)(a_1/b_1)$ where both factors are non units, then $a_i = pc_i$ for $i = 0, 1$. Hence, $pb_0b_1 = p^2c_0c_1$ which is impossible since the left hand side is only divisible by p once. Thus, p is irreducible, and so R is a DVR with uniformizing parameter $t = p$.

The last thing we need to show is that the quotient field of R, denoted by $\mathrm{Frac}(R)$, is $\mathbb{Q}$. If we recall that the quotient field of a ring is the smallest field that

contains it, then $\mathbb{Z} \subseteq R$ implies that $\mathbb{Q} = \mathrm{Frac}(\mathbb{Z}) \subseteq \mathrm{Frac}(R)$ and $R \subseteq \mathbb{Q}$ implies that $\mathrm{Frac}(R) \subseteq \mathbb{Q}$. Thus, $\mathrm{Frac}(R) = \mathbb{Q}$.

2.26. Let R be a DVR with quotient field K; let $\mathfrak{m}$ be the maximal ideal of R. (a) Show that if $z \in K$, $z \notin R$, then $z^{-1} \in \mathfrak{m}$. (b) Suppose $R \subset S \subset K$, and S is also a DVR. Suppose the maximal ideal of S contains $\mathfrak{m}$. Show that $S = R$.

Solution

(a) Since R is a DVR, then every non-zero element of K can be written as ut^n where $u \in R^\times$, t is a fixed uniformizing parameter for R, and $n \in \mathbb{Z}$. Hence, if $z \in K \setminus R$, then we must have that $z \neq 0$ and hence, that $z = ut^n$ with $n < 0$. It follows that $z^{-1} = u^{-1}t^{-n}$. Since $u^{-1} \in R^\times$ and $-n > 0$, then $z^{-1} = R \setminus R^\times = \mathfrak{m}$.

(b) First, let t be a uniformizing parameter for S and suppose that $t \notin R$, then $t \in S \setminus R \subset K \setminus R$. Thus, by part (a), $t^{-1} \in \mathfrak{m} \subset R \subset S$. But this is impossible since t is not a unit in S. Thus, $t \in R$.

Next, let's show that t is a uniformizing parameter for R. Since $t \in R$ and R is a DVR, then $t = ut_0^{n_0}$ where t_0 is a uniformizing parameter for R, $u \in R^\times \subset S^\times$ and $n_0 \geq 0$. Similarly, $t_0 \in R \setminus \{0\} \subset S \setminus \{0\}$ so $t_0 = vt^{n_1}$ where $v \in S^\times$ and $n_1 \geq 0$. It follows that $t = (uv^{n_0})t^{n_0 n_1}$. Viewing this equation as an equation in S, we get that $n_0 n_1 = 1$, and hence, $n_0 = 1$. Thus, $t = ut_0$ where $u \in R^\times$. Therefore, t is a uniformizing parameter for R.

Finally, let z be an arbitrary non-zero element in S, then $z = ut^n$ where u is a unit, and $n \geq 0$. Since $t \in R$, then $t^n \in R$. Suppose that $u \notin R$, then $u \in K \setminus R$, and hence, $u^{-1} \in \mathfrak{m}$. Since t is a uniformizing parameter in R and u^{-1} is not a unit in R, then $u^{-1} = vt^m$ for some unit $v \in R^\times$ and $m \geq 1$. Bt if we view this equation as an equation in S, we get that $m = 0$, a contradiction. So we must have $u \in R$ as well, and hence, $z = ut^n$. Therefore, $S = R$.

[**This proof never uses the assumption that the maximal ideal of S contains $\mathfrak{m}$. After rereading my proof, it seems like there is no mistake, and hence, the assumption seems to be unecessary.**]

2.27. Show that the DVR's of Problem 2.24 are the only DVR's with quotient field $k(X)$ that contain k. Show that those of Problem 2.25 are the only DVR's with quotient field $\mathbb{Q}$.

Solution **TODO**

2.28. An *order function* on a field K is a function φ from K onto $\mathbb{Z} \cup \{\infty\}$, satisfying:

(i) $\varphi(a) = \infty$ if and only if $a = 0$.

(ii) $\varphi(ab) = \varphi(a) + \varphi(b)$.

(iii) $\varphi(a + b) \geq \min(\varphi(a), \varphi(b))$.

Show that $R = \{z \in K | \varphi(z) \geq 0\}$ is a DVR with maximal ideal $\mathfrak{m} = \{z | \varphi(z) > 0\}$, and quotient field K. Conversely, show that if R is a DVR with quotient field K, then the function $\mathrm{ord} : K \to \mathbb{Z} \cup \{\infty\}$ is an order function on K. Giving a DVR with quotient field K is equivalent to defining an order function on K.

Solution **TODO**

2.29. **TODO**

Solution **TODO**

2.30. **TODO**

Solution **TODO**

2.31. **TODO**

Solution **TODO**

2.32. **TODO**

Solution **TODO**

2.6 Forms

2.33. **TODO**

Solution **TODO**

2.34. **TODO**

Solution **TODO**

2.35. **TODO**

Solution **TODO**

2.36. **TODO**

Solution **TODO**

2.7 Direct Products of Rings

2.37. **TODO**

Solution **TODO**

2.38. **TODO**

Solution **TODO**

2.8 Operations with Ideals

2.39. **TODO**

Solution **TODO**

2.40. **TODO**

Solution **TODO**

2.41. **TODO**

Solution **TODO**

2.42. **TODO**

Solution **TODO**

2.43. **TODO**

Solution **TODO**

2.44. **TODO**

Solution **TODO**

2.45. **TODO**

Solution **TODO**

2.46. **TODO**

Solution **TODO**

2.9 Ideals with a Finite Number of Zeros

2.47. TODO

Solution TODO

2.10 Quotient Modules and Exact Sequences

2.48. **TODO**

Solution **TODO**

2.49. **TODO**

Solution **TODO**

2.50. **TODO**

Solution **TODO**

2.51. **TODO**

Solution **TODO**

2.52. **TODO**

Solution **TODO**

2.53. **TODO**

Solution **TODO**

2.11 Free Modules

2.54. **TODO**

Solution **TODO**

2.55. **TODO**

Solution **TODO**

2.56. **TODO**

Solution **TODO**

Chapter 3

Local Properties of Plane Curves

3.1 Multiple Points and Tangent Lines

3.1. Prove that in the above examples $P = (0,0)$ is the only multiple point on the curves C, D, E, and F.

Solution

(a) Let $F = Y^2 - X^3$, then $F_X = -3X^2$ and $F_Y = 2Y$. A point $P = (a,b)$ is a multiple point if and only if $F_X(P) = F_Y(P) = 0$. This is equivalent to $-3a^2 = 2b = 0$, i.e., $P = (0,0)$. Therefore, the only multiple point is $P = (0,0)$.

(b) Let $F = Y^2 - X^3 - X^2$, then $F_X = -3X^2 - 2X$ and $F_Y = 2Y$. A point $P = (a,b) \in F$ is a multiple point if and only if $F_X(P) = F_Y(P) = 0$. This is equivalent to $-3a^2 - 2a = 2b = 0$. These equations are equivalent to $a = 0, -2/3$ and $b = 0$, but the point $(-2/3, 0) \notin F$ so the only solution is $P = (0,0)$. Therefore, the only multiple point is $P = (0,0)$.

(c) Let $F = (X^2 + Y^2)^2 + 3X^2Y - Y^3$, then $F_X = 4X^3 + 4XY^2 + 6XY$ and $F_Y = 4X^2Y + 4Y^3 + 3X^2 - 3Y^2$. A point $P = (a,b) \in F$ is a multiple point if and only if $F_X(P) = F_Y(P) = 0$. This is equivalent to the following system of equations:

$$\begin{cases} 4a^3 + 4ab^2 + 6ab = 0 \\ 4a^2b + 4b^3 + 3a^2 - 3b^2 = 0. \end{cases}$$

Clearly, $P = (0,0)$ is a solution of the system and a point in F so it is a multiple point. Moreover, by looking at the definition of F, we see that $(0,0)$ is the only point in F such that $X = 0$ or $Y = 0$. Hence, to find the other multiple points, we can assume that there is a multiple point (a,b) such that both a and b are non-zero. Dividing the first equation by a gives us

$$4a^2 + 4b^2 + 6b = 0$$

which is the same as

$$a^2 = -\frac{1}{2}(2b^2 + 3b).$$

Replacing a^2 in the second equation with the expression above gives us

$$b = -9/22$$

after making some simplifications. It follows that $a = 54/121$, and hence, we get the point $(54/121, -9/22)$. However, this point is not in F. Therefore, the only multiple point is $P = (0, 0)$.

(d) Let $F = (X^2 + Y^2)^3 - 4X^2Y^2$, then $F_X = 2X[3(X^2 + Y^2)^2 - 4Y^2]$ and $F_Y = 2Y[3(X^2 + Y^2)^2 - 4X^2]$. A point $P = (a, b) \in F$ is a multiple point if and only if $F_X(P) = F_Y(P) = 0$. This is equivalent to the following system of equations:

$$\begin{cases} 2a[3(a^2 + b^2)^2 - 4b^2] = 0 \\ 2b[3(a^2 + b^2)^2 - 4a^2] = 0. \end{cases}$$

Clearly, $P = (0, 0)$ is a solution of the system and a point in F so it is a multiple point. Moreover, by looking at the definition of F, we see that $(0, 0)$ is the only point in F such that $X = 0$ or $Y = 0$. Hence, to find the other multiple points, we can assume that there is a multiple point (a, b) such that both a and b are non-zero. Dividing the first equation by $2a$ and the second by $2b$ gives us

$$\begin{cases} 3(a^2 + b^2)^2 - 4b^2 = 0 \\ 3(a^2 + b^2)^2 - 4a^2 = 0. \end{cases}$$

It follows that $4b^2 = 4a^2$, and hence, that $a^2 = b^2$. Replacing b^2 with a^2 in the first equation of the latter system gives us the equation

$$3(2a^2)^2 - 4a^2 = 0$$

which is equivalent to $a = \pm 1/\sqrt{3}$. From $a^2 = b^2$, we get that $b = \pm 1/\sqrt{3}$ as well. However, $F(\pm 1/\sqrt{3}, \pm 1\sqrt{3}) \neq 0$ so the points $(\pm 1/\sqrt{3}, \pm 1/\sqrt{3})$ are not on the curve F. Therefore, the only multiple point is $P = (0, 0)$.

3.2. Find the multiple points, and the tangent lines at the multiple points, for each of the following curves:

(a) $Y^3 - Y^2 + X^3 - X^2 + 3XY^2 + 3X^2Y + 2XY$

(b) $X^4 + Y^4 - X^2Y^2$

(c) $X^3 + Y^3 - 3X^2 - 3Y^2 + 3XY + 1$

(d) $Y^2 + (X^2 - 5)(4X^4 - 20X^2 + 25)$

Sketch the part of the curve in (d) contained in $\mathbb{A}^2(\mathbb{R}) \subset \mathbb{A}^2(\mathbb{C})$.

Solution

(a) Let $F = Y^3 - Y^2 + X^3 - X^2 + 3XY^2 + 3X^2Y + 2XY$, then $F_X = 3X^2 - 2X + 3Y^2 + 6XY + 2Y$ and $F_Y = 3Y^2 - 2Y + 6XY + 3X^2 + 2X$. The system $F_X(a, b) = F_Y(a, b) = 0$ is equivalent to **TODO**

(b) **TODO**

(c) **TODO**

(d) **TODO**

3.3. **TODO**

Solution **TODO**

3.4. **TODO**

Solution **TODO**

3.5. **TODO**

Solution **TODO**

3.6. **TODO**

Solution **TODO**

3.7. **TODO**

Solution **TODO**

3.8. **TODO**

Solution **TODO**

3.9. **TODO**

Solution **TODO**

3.10. **TODO**

Solution **TODO**

3.11. **TODO**

Solution **TODO**

3.2 Multiplicities and Local Rings

3.12. **TODO**

Solution **TODO**

3.13. **TODO**

Solution **TODO**

3.14. **TODO**

Solution **TODO**

3.15. **TODO**

Solution **TODO**

3.16. **TODO**

Solution **TODO**

3.3 Intersection Numbers

3.17. **TODO**

Solution **TODO**

3.18. **TODO**

Solution **TODO**

3.19. **TODO**

Solution **TODO**

3.20. **TODO**

Solution **TODO**

3.21. **TODO**

Solution **TODO**

3.22. **TODO**

Solution **TODO**

3.23. **TODO**

Solution **TODO**

3.24. **TODO**

Solution **TODO**

Chapter 4

Projective Varieties

4.1 Projective Space

4.1. What points in $\mathbb{P}^2$ do not belong to two of the three sets U_1, U_2, U_3?

Solution By definition, a point $P = [x_1 : x_2 : x_3] \notin U_i$ if and only if $x_i = 0$. Hence, the point P do not belong to two of the three sets U_1, U_2, U_3 if and only if two of its three coordinates are zero. Thus, the possible values of P are $[x : 0 : 0]$, $[0 : y : 0]$, and $[0 : 0 : z]$ where $x, y, z \in k \setminus \{0\}$. Finally, these points are equal to the points $[1 : 0 : 0]$, $[0 : 1 : 0]$, and $[0 : 0 : 1]$. Therefore, these three points are precisely the three points that do not belong to two of the three sets U_1, U_2, U_3.

4.2. Let $F \in k[X_1, ..., X_{n+1}]$ (k infinite). Write $F = \sum F_i$, F_i a from of degree i. Let $P \in \mathbb{P}^n(k)$, and suppose $F(x_1, ..., x_{n+1}) = 0$ for every choice of homogeneous coordinates $(x_1, ..., x_{n+1})$ for P. Show that each $F_i(x_1, ..., x_{n+1}) = 0$ for all homogeneous coordinates for P. (*Hint:* consider $G(\lambda) = F(\lambda x_1, \ldots, \lambda x_{n+1}) = \sum \lambda^i F_i(x_1, ..., x_{n+1})$ for fixed $(x_1, ..., x_{n+1})$.)

Solution Let $x \in \mathbb{A}^{n+1}(k)$ be a choice of coordinates for the point P, and define the polynomial $G(\lambda) = F(\lambda x) = \lambda^i F_i(x)$. Here, since x is fixed, then $F_i(x)$ can be seen as the coefficients defining G. Since λx is another choice of coordinates for the point P for all $\lambda \neq 0$, then $G(\lambda) = F(\lambda x) = 0$ for all $\lambda \neq 0$. Since k is infinite, then G is infinitely many roots, and hence, $G = 0$. It follows that all the coefficients defining G are zero, giving us $F_i(x) = 0$. Since x was an arbitrary choice of coordinates for P, then $F_i(x_1, ..., x_{n+1}) = 0$ for all i and for all choice of coordinates for P.

4.3. (a) Show that the definitions of this section carry over without change to the case where k is an arbitrary field. (b) If k_0 is a subfield of k, show that $\mathbb{P}^n(k_0)$ may be identified with a subset of $\mathbb{P}^n(k)$.

Solution

(a) The only definitions in this section are the definitions of $\mathbb{P}^n(k)$, U_i and H_∞. None of these definitions used (explicitly or implicitly) properties not shared by all fields. Thus, they carry over without change to the case where k is an arbitrary field.

(b) Consider the function $i : \mathbb{P}^n(k_0) \to \mathbb{P}^n(k)$ defined by

$$i([x_1 : ... : x_{n+1}]_{\mathbb{P}^n(k_0)}) = [x_1 : ... : x_{n+1}]_{\mathbb{P}^n(k)}$$

where the subscript denotes the space in which the point lives in. First, let's show that this function is well-defined by showing that the image of a point doesn't depend on the choice of coordinates. Let $P = [x_1 : ... : x_{n+1}]_{\mathbb{P}^n(k_0)}$, then any other choice of coordinates for P would be of the form $[\lambda x_1 : ... : \lambda x_{n+1}]_{\mathbb{P}^n(k_0)}$ for some non-zero $\lambda \in k_0$. Hence:

$$\begin{aligned} i([\lambda x_1 : ... : \lambda x_{n+1}]_{\mathbb{P}^n(k_0)}) &= [\lambda x_1 : ... : \lambda x_{n+1}]_{\mathbb{P}^n(k)} \\ &= [x_1 : ... : x_{n+1}]_{\mathbb{P}^n(k)} \\ &= i([x_1 : ... : x_{n+1}]_{\mathbb{P}^n(k_0)}). \end{aligned}$$

Therefore, i is well-defined. Now, let's show that i is injective. Let $P = [x_1 : ... : x_{n+1}]_{\mathbb{P}^n(k_0)}$ and $Q = [y_1 : ... : y_{n+1}]_{\mathbb{P}^n(k_0)}$ such that $i(P) = i(Q)$, then equivalently:

$$[x_1 : ... : x_{n+1}]_{\mathbb{P}^n(k)} = [y_1 : ... : y_{n+1}]_{\mathbb{P}^n(k)}.$$

This implies that $y_i = \lambda x_i$ for all i and for some non-zero constant $\lambda \in k$. Let i_0 be such that $x_{i_0} \neq 0$, then it follows that $\lambda = y_{i_0}/x_{i_0} \in k_0$. Thus, $y_i = \lambda x_i$ for all i and for some non-zero constant $\lambda \in k_0$. It follows that

$$P = [x_1 : ... : x_{n+1}]_{\mathbb{P}^n(k_0)} = [y_1 : ... : y_{n+1}]_{\mathbb{P}^n(k_0)} = Q.$$

Therefore, i is injective, and hence, we can naturally identify $\mathbb{P}^n(k_0)$ with a subset of $\mathbb{P}^n(k)$ (the image of i).

4.2 Projective Algebraic Sets

4.4. Let I be a homogeneous ideal in $k[X_1, ..., X_{n+1}]$. Show that I is prime if and only if the following condition is satisfied; for any forms $FG \in k[X_1, ..., X_{n+1}]$, if $FG \in I$, then $F \in I$ or $G \in I$.

Solution If I is prime, then the condition is clearly satisfied. Suppose now that I satisfies the given condition and let $F, G \in k[X_1, ..., X_{n+1}]$ such that $F = \sum F_i$, $G = \sum G_i$ and $FG \in I$. Since

$$FG = F_0G_0 + (F_0G_1 + F_1G_0) + (F_0G_2 + F_1G_1 + F_2G_0) + ...,$$

and I is homogeneous, then each term in the sum above is in I. Hence, $F_0G_0 \in I$. Since I satisfies the given condition, then $F_0 \in I$ or $G_0 \in I$. If $F_0 \notin I$, then $G_0 \in I$, and so

$$F_0G_1 = (F_0G_1 + F_1G_0) - F_1G_0 \in I.$$

This again implies that $G_1 \in I$ since $F_0 \in I$. By induction, we can show that this case implies that $G_i \in I$ for all i, and hence, that $G \in I$. Similarly, if $F_0 \in I$, then

$$F_1G_0 = (F_0G_1 + F_1G_0) - F_0G_1 \in I$$

and hence, $F_1 \in I$ or $G_0 \in I$. by repeating this procedure, we can either fall in the case where all the F_i's are in I, and hence deduce that $F \in I$, or, there will be one F_i which is not in I, and hence, forcing all the G_i's to be in I, which would imply that $G \in I$. Thus, in all possible cases, $F \in I$ or $G \in I$. Therefore, I is prime.

4.5. If I is a homogeneous ideal, show that $\mathrm{Rad}(I)$ is also homogeneous.

Solution Let $F \in \mathrm{Rad}(I)$ such that $F = \sum_{i=m}^{d} F_i$ where $\deg(F_i) = i$, then $F^n \in I$ for some $n \geq 1$. It follows that

$$F^n = F_m^n + [\text{terms of higher degree}] \in I$$

Since I is homogeneous and F_m^n is the form of smallest degree in the decomposition of F^n into a sum of forms, then $F_m^n \in I$. Hence, $F_m \in \mathrm{Rad}(I)$, and so $F - F_m \in \mathrm{Rad}(I)$. It suffices to repeat this argument (induction) to get that $F_i \in \mathrm{Rad}(I)$ for all i. Therefore, $\mathrm{Rad}(I)$ is homogeneous.

4.6. State and prove the projective analogues of properties (1)-(10) of Chapter 1, Sections 2 and 3.

Solution

(1) Let's show that if I is the ideal in $k[X_1, ..., X_{n+1}]$ generated by the subset $S \subset k[X_1, ..., X_{n+1}]$, then $V(S) = V(I)$. Let $P \in V(I)$, then $F(P) = 0$ for all $F \in I$. Since $S \subset I$, then $F(P) = 0$ for all $F \in S$. Hence, $V(I) \subset V(S)$. Conversely, let $P \in V(S)$, then $F(P) = 0$ for all $F \in S$. Let $G \in I$, then $G = \sum \alpha_i F_i$ for some $\alpha_i \in k[X_1, ..., X_{n+1}]$ and $F_i \in S$. Let $(x_1, ..., x_{n+1})$ be a choice of homogeneous coordinates for P, then

$$G(x_1, ..., x_{n+1}) = \sum \alpha_i(x_1, ..., x_{n+1}) F_i(x_1, ..., x_{n+1}) = 0.$$

Since it holds for all choice of homogeneous coordinates for P, then $G(P) = 0$. Since it holds for all $G \in I$, then $P \in V(I)$. Therefore, $V(S) = V(I)$.

(2) Let's show that if $\{I_\alpha\}$ is any collection of ideals, then $V(\bigcup_\alpha I_\alpha) = \bigcap_\alpha V(I_\alpha)$. Let $P \in V(\bigcup_\alpha I_\alpha)$, then $F(P) = 0$ for all $F \in \bigcup_\alpha I_\alpha$. Let α be arbitrary and $F \in I_\alpha$, then $F \in \bigcup_\alpha I_\alpha$, and hence, $F(P) = 0$. Thus, $P \in V(I_\alpha)$. Since α was set to be arbitrary, then $P \in \bigcap_\alpha V(I_\alpha)$. Conversely, let $P \in \bigcap_\alpha V(I_\alpha)$ and take any $F \in \bigcup_\alpha I_\alpha$, then $F \in I_{\alpha'}$ for some α'. Since $P \in V(I_{\alpha'})$, then $F(P) = 0$. Since it holds for all $F \in \bigcup_\alpha I_\alpha$, then $P \in V(\bigcup_\alpha I_\alpha)$. Therefore, $V(\bigcup_\alpha I_\alpha) = \bigcap_\alpha V(I_\alpha)$.

(3) Let's show that if $I \subset J$, then $V(I) \supset V(J)$. Let $P \in V(J)$, then $F(P) = 0$ for all $F \in J$. Let $F \in I$, then $F \in J$, and hence, $F(P) = 0$. Since it holds for all $F \in I$, then $P \in V(I)$. Therefore, $V(I) \supset V(J)$.

(4) Let's show that $V(FG) = V(F) \cup V(G)$, and that $V(I) \cup V(J) = V(\{FG | F \in I, G \in J\})$. Let $P \in V(FG)$, then $(FG)(P) = 0$. Hence, $F(x_1, ..., x_{n+1}) = 0$ or $G(x_1, ..., x_{n+1}) = 0$ for all choice of homogeneous coordinates for P. Fix a choice of homogeneous coordinates $P = [x_1 : ... : x_{n+1}]$ and define the polynomials

$$f(\lambda) = F(\lambda x_1, ..., \lambda x_{n+1}), \qquad g(\lambda) = G(\lambda x_1, ..., \lambda x_{n+1}).$$

By our previous remark, for all non-zero $\lambda \in k$, either $f(\lambda) = 0$ or $g(\lambda) = 0$. Since k is infinite, then it follows that either f or g has infinitely many zeros, without loss of generality, suppose that f has infinitely many zeros. Since f is a one-variable polynomial, then we must have $f = 0$. Equivalently, this means that $F(P) = 0$. Thus, $P \in V(F) \subset V(F) \cup V(G)$. Conversely, suppose that $P \in V(F) \cup V(G)$ and suppose that $P \in V(F)$ without loss of generality, then $F(P) = 0$. Equivalently, this means that $F(x_1, ..., x_{n+1}) = 0$ for all choice of homogeneous coordinates for P. Hence, $F(x_1, ..., x_{n+1})G(x_1, ..., x_{n+1}) = 0$ for all choice of coordinates for P. By definition, this implies that $(FG)(P) = 0$, and hence, that $P \in V(FG)$. Therefore, $V(FG) = V(F) \cup V(G)$.

Let's now prove the more general statement using what we just proved. Let $P \in V(\{FG | F \in I, G \in J\})$, then $(FG)(P) = 0$ for all $F \in I$ and $G \in J$. By the previous proof, it follows that $F(P) = 0$, or $G(P) = 0$. Now, suppose that $P \in V(I)$, then we are done. Otherwise, there is a $F_0 \in I$ such that $F_0(P) \neq 0$. Thus, for all $G \in J$, the equation $(F_0G)(P) = 0$ implies that $F_0(P) = 0$ or $G(P) = 0$, which implies that $G(P) = 0$. Thus, in that case, $P \in V(J)$. Thus, in both cases, $P \in V(I) \cup V(J)$. Conversely, let $P \in V(I) \cup V(J)$ and suppose without loss of generality that $P \in V(I)$, then $F(P) = 0$ for all $F \in I$. It follows that $(FG)(P) = 0$ for all $F \in I$ and $G \in J$. Therefore, $P \in V(\{FG | F \in I, G \in J\})$. Therefore, $V(I) \cup V(J) = V(\{FG | F \in I, G \in J\})$.

(5) Let's show that $V(0) = \mathbb{P}^n$; $V(1) = \varnothing$; $\{[x_1 : ... : x_{n+1}]\}$ is a projective algebraic set for all $P \in \mathbb{P}^n$. First, we clearly have that every point in $\mathbb{P}^n$ is a zero of the zero polynomial so $V(0) = \mathbb{P}^n$. Similarly, no point in $\mathbb{P}^n$ is a zero of the polynomial 1 so $V(1) = \varnothing$. Next, let P be a point in $\mathbb{P}^n$ and fix some homogeneous coordinates $(x_1, ..., x_{n+1})$ for P. In $\mathbb{A}^{n+1}(k)$, the line L

through the origin and $(x_1, ..., x_{n+1})$ is an algebraic set, so it can be described as $V(f_1, ..., f_r) \subset \mathbb{A}^{n+1}(k)$ for some polynomials $f_i \in k[x_1, ..., x_{n+1}]$ (Problem 2.15). For all $\lambda \neq 0$, the point $(\lambda x_1, ..., \lambda x_{n+1})$ is in the line, and hence, it is in $V(f_1, ..., f_r)$. Thus, $f_i(\lambda x_1, ..., \lambda x_{n+1}) = 0$ for all i and for all $\lambda \neq 0$. By definition, this means that P is a zero for all f_i, so $\{P\} \subset V(f_1, ..., f_r)$ viewed now as a subset of $\mathbb{P}^n$. Conversely, let $Q \in \mathbb{P}^n$ such that $Q \in V(f_1, ..., f_r)$, then choosing some coordinates $(y_1, ..., y_{n+1})$ for Q, we get that $(y_1, ..., y_{n+1}) \in V(f_1, ..., f_r) = L \subset \mathbb{A}^{n+1}(k)$. Hence, $(y_1, ..., y_{n+1}) = (\lambda x_1, ..., \lambda x_{n+1})$ for some $\lambda \neq 0$ so $Q = P$. Therefore, $V(f_1, ..., f_r) = \{P\}$ where $V(f_1, ..., f_r)$ is now viewed as a subset of $\mathbb{P}^n$.

(6) Let's prove that if $X \subset Y$, then $I(X) \supset I(Y)$. Let $F \in I(Y)$, then $F(P) = 0$ for all $P \in Y$. Let $P \in X$, then $P \in Y$, and hence, $F(P) = 0$. Since it holds for all $P \in X$, then $F \in I(X)$. Therefore, $I(X) \supset I(Y)$.

(7) Let's show that $I(\varnothing) = k[X_1, ..., X_{n+1}]$ and $I(\mathbb{P}^n(k)) = (0)$. First, for all $F \in k[X_1, ..., X_{n+1}]$, all the points in $\varnothing$ are zeros of F, so $F \in I(\varnothing)$. Therefore, $I(\varnothing) = k[X_1, ..., X_{n+1}]$. Next, if $F \in I(\mathbb{P}^n(k))$, then $F(\mathbb{A}^{n+1}(k) \setminus \{0\}) = \{0\}$. If we define $f(\lambda) = F(\lambda, \ldots, \lambda)$, we get that f is zero for all $\lambda \neq 0$. Since f is a one-variable polynomial and k is infinite, then $f = 0$, and so $F(0) = 0$. Therefore, $F \in I(\mathbb{A}^{n+1}(k))$, and hence, $F = 0$. It follows that $I(\mathbb{P}^n(k)) = (0)$.

(8) Let's show that $I(V(S)) \supset S$ for any set S of polynomials; $V(I(X)) \supset X$ for any set X of points. Let $F \in S$ and $P \in V(S)$, then by definition of $V(S)$, $F(P) = 0$. Since it holds for all $P \in V(S)$, then $F \in I(V(S))$. Therefore, $I(V(S)) \supset S$. Next, Let $P \in X$ and $F \in I(X)$, then by definition, $F(P) = 0$. Since it holds for all $F \in I(X)$, then $P \in V(I(X))$. Therefore, $V(I(X)) \supset X$.

(9) Let's show that

$$V(I(V(S))) = V(S) \quad \text{and} \quad I(V(I(X))) = I(X)$$

for any set S of polynomials, and any set X of points. First, we know that $V(I(V(S))) \supset V(S)$ and $I(V(S)) \supset S$ from property (8). The inclusion $I(V(S)) \supset S$ implies $V(I(V(S))) \subset V(S)$ using property (3). Therefore, $V(I(V(S))) = V(S)$. Next, we have the inclusions $I(V(I(X))) \supset I(X)$ and $V(I(X)) \supset X$ from property (8). The latter implies that $I(V(I(X))) \subset I(X)$ using property (6). Therefore, $I(V(I(X))) = I(X)$.

(10) Let's show that $I(X)$ is a radical ideal for any $X \subset \mathbb{P}^n(k)$. Let $F^n \in I(X)$, then $F^n(P) = 0$ for all $P \in X$. By property (4), it follows that $F(P) = 0$ or $F^{n-1}(P) = 0$. If $F^{n-1}(P) = 0$, simply reapply the same procedure to get that in all cases, $F(P) = 0$. Since it holds for all $P \in X$, then $F \in I(X)$. Therefore, $I(X)$ is a radical ideal.

4.7. Show that each irreducible component of a cone is also a cone.

Solution From the equation $C(V_p(I)) = V_a(I)$, we get that the cone of a projective algebraic subset of $\mathbb{P}^n(k)$ is an affine algebraic set of $\mathbb{A}^{n+1}(k)$. Morever, it is easy to show that $C(V_1 \cup V_2) = C(V_1) \cup C(V_1)$. Hence, if we take a projective algebraic set V

such that $V = V_1 \cup \cdots \cup V_r$ is its unique decomposition into irreducible components, then $C(V) = C(V_1) \cup \cdots \cup C(V_r)$. Let's show that $C(V_1), ..., C(V_r)$ are the irreducible components of $C(V)$. First, they are all irreducible because $I_a(C(V_i)) = I_p(V_i)$ is prime. Morever, suppose that $C(V_i) \subset C(V_j)$, then $I_p(V_i) \subset I_p(V_j)$, and hence, $V_i \subset V_j$. But this is a contradiction since V_i and V_j are irreducible components of V. Therefore, $C(V_1), ..., C(V_r)$ are the irreducible components of $C(V)$. By uniqueness of the decomposition into irreducible components, we get that each irreducible component of a cone is also a cone.

4.8. Let $V = \mathbb{P}^1$, $\Gamma_h(V) = k[X, Y]$. Let $t = X/Y \in k_h(V)$, and show that $k(V) = k(t)$. Show that there is a natural one-to-one correspondence between the points of $\mathbb{P}^1$ and the DVR's with quotient field $k(V)$ that contain k (see Problem 2.27); which DVR corresponds to the point at infinity?

Solution Let f and g be two forms in $\Gamma_h(V)$ of the same degree, say d, then we can write $f = \sum_{i=0}^{d} a_i X^{d-i} Y^i$ and $g = \sum_{i=0}^{d} b_i X^{n-i} Y^i$ where $a_i, b_i \in k$, then

$$\begin{aligned}
\frac{f}{g} &= \frac{\sum_{i=0}^{d} a_i X^{d-i} Y^i}{\sum_{i=0}^{d} b_i X^{d-i} Y^i} \\
&= \sum_{i=0}^{d} a_i \frac{X^{d-i} Y^i}{\sum_{j=0}^{d} b_j X^{d-j} Y^j} \\
&= \sum_{i=0}^{d} a_i \left(\frac{\sum_{j=0}^{d} b_j X^{d-j} Y^j}{X^{d-i} Y^i} \right)^{-1} \\
&= \sum_{i=0}^{d} a_i \left(\sum_{j=0}^{d} b_j \frac{X^{d-j} Y^j}{X^{d-i} Y^i} \right)^{-1} \\
&= \sum_{i=0}^{d} a_i \left(\sum_{j=0}^{d} b_j \frac{X^{i-j}}{Y^{i-j}} \right)^{-1} \\
&= \sum_{i=0}^{d} a_i \left(\sum_{j=0}^{d} b_j t^{i-j} \right)^{-1} \in k(t).
\end{aligned}$$

Hence, $k(V) \subset k(t)$. Conversely, since k and t are in $k(V)$ and $k(V)$ is a field, then $k(t) \subset k(V)$. Therefore, $k(V) = k(t)$.

Next, since $k(V) = k(t)$, then the DVR's with quotient field $k(V)$ that contain k are precisely $\mathcal{O}_a$ for $a \in \mathbb{A}^1(k)$ with uniformizing parameter $t = X - a$, and $\mathcal{O}_\infty$ with uniformizing parameter $t = 1/X$. Associate each point $[a : 1] \in \mathbb{P}$ to $\mathcal{O}_a$ and the point at infinity $[0 : 1]$ to $\mathcal{O}_\infty$.

4.9. Let I be a homogeneous ideal in $k[X_1, ..., X_{n+1}]$, and

$$\Gamma = k[X_1, ..., X_{n+1}]/I.$$

Show that the forms of degree d in Γ form a finite-dimensional vector space over k.

Solution TODO

4.10. TODO

Solution TODO

4.11. TODO

Solution TODO

4.12. Let $H_1, ..., H_m$ be hyperplanes in $\mathbb{P}^n$, $m \leq n$. Show that $H_1 \cap H_2 \cap ... \cap H_m \neq \varnothing$.

Solution Since H_i is a hyperplane, then $H_i = V(f_i)$ where $f_i \in k[X_1, ..., X_{n+1}]$ is a form of degree. Hence, $\bigcap_i H_i = V(f_1, ..., f_m)$. An element $P \in \mathbb{P}^n$ is an element of $\bigcap_i H_i$ if and only if $f_i(P) = 0$ for all $1 \leq i \leq m$. If we write $f_i = a_{i1}X_1 + a_{i2}X_2 + \cdots + a_{i(n+1)}X_{n+1}$, then the condition is equivalent to the solvability of the following system of equations:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1(n+1)}x_{n+1} &= 0, \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2(n+1)}x_{n+1} &= 0, \\ \vdots \qquad\qquad\qquad \vdots \qquad &\quad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{m(n+1)}x_{n+1} &= 0. \end{aligned}$$

Define the linear transformation $T : k^{n+1} \to k^m$ as follows:

$$T(\vec{x}) = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1(n+1)} \\ a_{21} & a_{22} & \dots & a_{2(n+1)} \\ \dots & \dots & \ddots & \dots \\ a_{m1} & a_{m2} & \dots & a_{m(n+1)} \end{pmatrix} \vec{x},$$

then the system of equation is equivalent to the equation $T\vec{x} = 0$ with $x \in k^{n+1}$. By the rank-nullity theorem, we have that

$$\dim \ker T + \dim \operatorname{im} T = n + 1.$$

Since $\operatorname{im} T \subset k^m$, then

$$\dim \ker T = (n+1) - \dim \operatorname{im} T \geq (n+1) - m \geq 1.$$

Thus, there is at least one non-zero vector $(x_1, ..., x_{n+1})$ such that all of its scalar multiples are solution to the equation (by linearity). Hence, if we define $P = [x_1 : ... : x_{n+1}] \in \mathbb{P}^n$, then another way to phrase the previous sentence is to say that $f_i(P) = 0$ for all i, or equivalently, that $P \in \bigcap_i H_i$. Therefore, $H_1 \cap H_2 \cap ... \cap H_m \neq \varnothing$.

4.13. TODO

Solution TODO

4.14. **TODO**

Solution **TODO**

4.15. **TODO**

Solution **TODO**

4.16. **TODO**

Solution **TODO**

4.17. **TODO**

Solution **TODO**

4.18. **TODO**

Solution **TODO**

4.3 Affine and Projective Varieties

4.19. If $I = (F)$ is the ideal of an affine hypersurface, show that $I^* = (F^*)$.

Solution Since $F \in I$, then $F^* \in I^*$ and hence, $(F^*) \subset I^*$. Conversely, let $hF \in (F)$, then $(hF)^*$ is one of the generators of I^*. But notice that $(hF)^* = h^*F^* \in (F^*)$ so all the generators of I^* are in (F^*). Therefore, $I^* \subset (F^*)$, and hence, $I^* = (F^*)$.

4.20. Let $V = V(Y - X^2, Z - X^3) \subset \mathbb{A}^3$. Prove:

(a) $I(V) = (Y - X^2, Z - X^3)$.

(b) $ZW - XY \in I(V)^* \subset k[X, Y, Z, W]$, but $ZW - XY \notin ((Y - X^2)^*, (Z - X^3)^*)$.

So if $I(V) = (F_1, ..., F_r)$, it does not follow that $I(V)^* = (F_1^*, ..., F_r^*)$.

Solution

(a) The ideal $(Y - X^2, Z - X^3)$ is prime because $k[X, Y, Z]/(Y - X^2, Z - X^3) \cong k[X]$. Hence, by the Nullstellenzats, $I(V) = (Y - X^2, Z - X^3)$.

(b) Since $Y - X^2, Z - X^3 \in I(V)$, then

$$Z - XY = (Z - X^3) - X(Y - X^2) \in I(V).$$

It follows that

$$WZ - XY = (Z - XY)^* \in I(V)^*.$$

Next, suppose $WZ - XY \in ((Y - X^2)^*, (Z - X^3)^*) = (WY - X^2, W^2Z - X^3)$, then there exist polynomials $A, B \in k[X, Y, Z, W]$ such that

$$WZ - XY = A(X, Y, Z, W)(WY - X^2) + B(X, Y, Z, W)(W^2Z - X^3).$$

Taking $X = Y = 0$ gives us

$$WZ = B(0, 0, Z, W)W^2Z$$

which is equivalent to

$$1 = B(0, 0, Z, W)W.$$

But this last equation is impossible so $WZ - XY \notin ((Y - X^2)^*, (Z - X^3)^*)$.

4.21. Show that if $V \subset W \subset \mathbb{P}^n$ are varieties, and V is a hypersurface, then $W = V$ or $W = \mathbb{P}^n$ (see Problem 1.30).

Solution Since V is a hypersurface and a variety, then there is an irreducible polynomial $f \in k[X_1, ..., X_{n+1}]$ such that $V(f)$. From the inclusion $V \subset W$, we get $I(W) \subset I(V(f)) = (f)$ where the last equality comes from the Nullstellenzats. Since W is a variety, then $I(W)$ must also be a prime ideal. Suppose that $I(W)$ is not the zero ideal, then any non-zero element $h_0 \in I(W)$ is of the form $h_0 = h_1 f$ where $h_1 \in k[X_1, ..., X_{n+1}]$. Since $I(W)$ is prime, then $h_1 \in I(W)$ or $f \in I(W)$. If $f \notin I(W)$, then we can construct the infinite sequence $h_n \in I(W)$ such that

$h_0 = h_n f^n$. But this is impossible because it would mean that the degree of h is infinite. Therefore, $f \in I(W)$, and hence, $I(V) = (f) \subset I(W)$. In that case, $I(W) = I(V)$. In other words, either $I(W) = (0)$, or $I(W) = I(V)$. In the first case, $W = \mathbb{P}^n$, and in the second case, $W = V$.

4.22. Suppose V is a variety in $\mathbb{P}^n$ and $V \supset H_\infty$. Show that $V = \mathbb{P}^n$ or $V = H_\infty$. If $V = \mathbb{P}^n$, $V_* = \mathbb{A}^n$, while if $V = H_\infty$, $V_* = \varnothing$.

Solution First, notice that we can write H_∞ as $V(X_{n+1})$, which now makes it clear that H_∞ is an irreducible hypersurface. Hence, from the inclusion $H_\infty \subset V \subset \mathbb{P}^n$, we get $V = \mathbb{P}^N$ or $V = H_\infty$ using Problem 4.21. If $V = \mathbb{P}^n$, then $I(V) = (0)$, so $I(V)_* = (0)$, and hence, $V_* = V(I(V)_*) = V(0) = \mathbb{A}^n$. Similarly, if $V = H_\infty$, then $I(V) = (X_{n+1})$, so $I(V)_* = (1)$, and hence, $V_* = V(I(V)_*) = V(1) = \varnothing$.

4.23. Describe all subvarieties in $\mathbb{P}^1$ and in $\mathbb{P}^2$.

Solution First, we know that all subvarieties of $\mathbb{A}^n$ can be lifted to subvarieties of $\mathbb{P}^n$ through the correspondence $V \mapsto V^*$. The subvarieties of $\mathbb{A}^1$ are the empty set, the singeltons and $\mathbb{A}^1$. It follows that the empty set, the singeltons of U_2 and $\mathbb{P}^n$ are subvarieties of $\mathbb{P}^n$. Since all singeltons are subvarieties, then we can add to this list the point at infinity. Finally, any subvariety that is contained in H_∞ is either $\varnothing$ or H_∞ (since H_∞ is a singelton in this case) and any subvariety that contains H_∞ is either equal to H_∞ or $\mathbb{P}^n$ by the previous problem. Therefore, the complete list of subvarieties of $\mathbb{P}^1$ is: $\varnothing$, $\{P\}$ with $P \in \mathbb{P}^n$, and $\mathbb{P}^n$.

Let's find the subvarieties of $\mathbb{P}^2$ in a similar way. First, recall that the subvarieties of $\mathbb{A}^2$ are: $\varnothing$, points, $V(f)$ where f is irreducible and non-constant, and $\mathbb{A}^2$. It follows that $\varnothing$, points (including the points in H_∞ since we already know that all singeltons are algebraic sets), $V(f^*)$ where f is an irreducible non-constant polynomial, and $\mathbb{P}^n$ are subvarieties of $\mathbb{P}^2$. The last potential subvarieties of $\mathbb{P}^2$ are the ones that are either contained in H_∞, or that contains H_∞. In the first case, by the correspondence between H_∞ and $\mathbb{P}^1$, we get that these varieties must be the empty set, the points in H_∞ or H_∞ entirely. In the second case, the subvarieties must be either H_∞ or $\mathbb{P}^n$ (using Problem 4.22 again). Thus, in our list of subvarieties of $\mathbb{P}^n$, we only need to add H_∞ to complete it. Therefore, the complete list of subvarieties of $\mathbb{P}^2$ is: $\varnothing$, all points, $V(f^*)$ where f is an irreducible non-constant polynomial, H_∞, and $\mathbb{P}^n$.

4.24. **TODO**

Solution **TODO**

4.25. **TODO**

Solution **TODO**

4.4 Multiprojective Space

4.26. **TODO**

Solution **TODO**

4.27. **TODO**

Solution **TODO**

4.28. **TODO**

Solution **TODO**

Chapter 5

Projective Plane Curves

TODO

Chapter 6

Varieties, Morphisms, and Rational Maps

6.1 The Zariski Topology

6.1. Let $Z \subset Y \subset X$, X a topological space. Give Y the induced topology. Show that the topology induced by Y on Z is the same as that induced by X on Z.

Solution Let $U \subset Z$ be an open set from the topology induced by Y, then there is an open subset V of Y such that $U = Z \cap V$. Similarly, since V is from the topology induced by X on Y, there is an open subset W of X such that $V = Y \cap W$. Hence, $U = Z \cap Y \cap W = Z \cap W$, from which it follows that U is also open in the sense of the topology induced by X on Z.

Conversely, let $U \subset Z$ be an open set from the topology induced by X on Z, then there is an open subset V of X such that $U = Z \cap V$. Thus, we can write $U = Z \cap (Y \cap V)$. Since $Y \cap V$ is an open subset of Y from the topology induced by X on Y, then $U = Z \cap (Y \cap V)$ is an open subset from the topology induced by Y on Z. Therefore both topologies are the same.

6.2. (a) Let X be a topological space, $X = \bigcup_{\alpha \in \mathcal{A}} U_\alpha$, U_α open in X. Show that a subset W of X is closed if and only if each $W \cap U_\alpha$ is closed (in the induced topology) in U_α. (b) Suppose similarly $Y = \bigcup_{\alpha \in \mathcal{A}} V_\alpha$, V_α open in Y, and suppose $f : X \to Y$ is mapping such that $f(U_\alpha) \subset V_\alpha$. Show that f is continuous if and only if the restriction of f to each U_α is a continuous mapping from U_α to V_α.

Solution

(a) Suppose W is closed, then from the equation

$$U_\alpha \setminus (W \cap U_\alpha) = U_\alpha \cap (X \setminus W),$$

we get that $W \cap U_\alpha$ must be closed in U_α for all α. Conversely, suppose that $W \cap U_\alpha$ is closed in U_α for all α, then by the previous equation $U_\alpha \cap (X \setminus W)$ is an open subset of X for all α. It follows that

$$\bigcup_\alpha [U_\alpha \cap (X \setminus W)] = (X \setminus W) \cap \bigcup_\alpha U_\alpha = (X \setminus W) \cap X = X \setminus W$$

is open in X. Therefore, W is a closed subset of X.

(b) Suppose that f is continuous and let U be an open subset of V_α, then $U = V_\alpha \cap V$ where V is an open subset of Y. It follows that

$$f|_{U_\alpha}^{-1}(U) = U_\alpha \cap f^{-1}(V_\alpha \cap V).$$

Since both V_α and V are open subsets of Y, then $V_\alpha \cap V$ is also an open subset of Y. Thus by continuity of f, the set $f|_{U_\alpha}^{-1}(U)$ is open in U_α since $f^{-1}(V_\alpha \cap V)$ is open in X. Therefore, $f|_{U_\alpha} : U_\alpha \to V_\alpha$ is continuous for all α. Conversely, suppose that f_{U_α} is continuous for all α and let U be an open subset of Y, then

$$f^{-1}(U) = \bigcup_\alpha U_\alpha \cap f^{-1}(U) = \bigcup_\alpha f|_{U_\alpha}^{-1}(U \cap V_\alpha).$$

By our assumption, $f|_{U_\alpha}^{-1}(U \cap V_\alpha)$ must be open since $U \cap V_\alpha$ is open in V_α and $f|_{U_\alpha}$ is continuous. Thus, $f^{-1}(U)$ is open. Therefore, f is continuous on X.

6.3. (a) Let V be an affine variety, $f \in \Gamma(V)$. Considering f as a mapping from V to $k = \mathbb{A}^1$, show that f is continuous. (b) Show that any polynomial map of affine varieties is continuous.

Solution

(a) We can suppose that f is the restriction of a polynomial to V. Recall that the algebraic subsets of $\mathbb{A}^1$ are the empty set, the finite sets, and $\mathbb{A}^1$. Let's show that in each case, $f^{-1}(C)$ is a closed subset of V where C is a closed subset of $\mathbb{A}^1$. Trivially, $f^{-1}(\varnothing) = \varnothing$ and $f^{-1}(\mathbb{A}^1) = V$, which are both closed subsets of V. Finally, let $C = \{a_1, ..., a_r\}$ be a finite subset of $\mathbb{A}^1$, then

$$f^{-1}(C) = V \cap (V(f - a_1) \cup V(f - a_2) \cup ... \cup V(f - a_r)).$$

Since each $V(f - a_i)$ is closed in $\mathbb{A}^n$, then $V(f - a_1) \cup ... \cup V(f - a_r)$ is a closed subset of $\mathbb{A}^n$. It follows that $f^{-1}(C)$ is a closed subset of V. Therefore, f is a continuous function from V to $k = \mathbb{A}^1$.

(b) Let $\varphi : V \to W$ be a polynomial map and let $C = W \cap V(f_1, ..., f_r)$ be an arbitrary closed subset of W, then

$$\varphi^{-1}(C) = V \cap V(f_1 \circ \varphi, ..., f_r \circ \varphi)$$

which is a closed subset of V. therefore, polynomial maps are continuous.

6.4. Let $U_i \subset \mathbb{P}^n$, $\varphi_i : \mathbb{A}^n \to U_i$ as in Chapter 4. Give U_i the topology induced from $\mathbb{P}^n$. (a) Show that φ_i is a homeomorphism. (b) Show that a set $W \subset \mathbb{P}^n$ if and only if each $\varphi_i^{-1}(W)$ is closed in $\mathbb{A}^n$, $i = 1, ..., n + 1$. (c) Show that if $V \subset \mathbb{A}^n$ is an affine variety, then the projective closure V^* of V is the closure of $\varphi_{n+1}(V)$ in $\mathbb{P}^n$.

Solution

(a) **TODO**

(b) **TODO**

(c) **TODO**

6.5. **TODO**

Solution **TODO**

6.6. **TODO**

Solution **TODO**

6.7. Let V be an affine variety, $f \in \Gamma(V)$. (a) Show that $V(f) = \{P \in V | f(P) = 0\}$ is a closed subset of V, and $V(f) \neq V$ unless $f = 0$. (b) Suppose U is a dense subset of V and $f(P) = 0$ for all $P \in U$. Then $f = 0$.

Solution

(a) Let F be a polynomial such that $\overline{F} = f$, then $V(f) = V \cap V(F)$, which is closed in V. Next, suppose $V(f) = V$, then $F(P) = 0$ for all $P \in V$. Hence, $F \in I(V)$ which implies that $f = \overline{F} = 0$ in $\Gamma(V)$.

(b) First, we can restate the statement of the question by writing that $U \subset V(f)$. Moreover, $V(f)$ is a closed set so the closure of U is also a subset of U. But U is dense so its closure is V. Hence, $V \subset V(f)$, so $V(f) = V$. Therefore, $f = 0$.

6.8. **TODO**

Solution **TODO**

6.2 Varieties

6.9. Let $X = \mathbb{A}^2 \setminus \{(0,0)\}$, an open subvariety of $\mathbb{A}^2$. Show that $\Gamma(X) = \Gamma(\mathbb{A}^2) = k[X,Y]$.

Solution Clearly, we have that $k[X,Y] \subset \Gamma(X)$ since polynomials are defined everywhere on X. Conversely, let $z \in \Gamma(X)$, then $z \in k(X,Y)$ and it is defined everywhere except maybe at zero. Since $k[X,Y]$ is a UFD, then we can write $z = a/b$ in a unique way (up to units) with $a, b \in k[X,Y]$. Hence, we get that $b(P) \neq 0$ for all $P \neq 0$. Let's prove that this implies that $b(0,0) \neq 0$ as well. **TODO**[Prouver que $b \equiv c$ sur $k^2 \setminus \{(0,0)\}$, donc $b - c$ est zero sur un sous ensemble ouvert de k^2, donc dense. Donc, b est zero sur k^2.]**TODO**

6.10. Let U be an open subvariety of a variety X, Y a closed subvariety of U. Let Z be the closure of Y in X. Show that

(a) Z is a closed subvariety of X.

(b) Y is an open subvariety of Z.

Solution

(a) **TODO**

(b) **TODO**

6.11. **TODO**

Solution **TODO**

6.12. **TODO**

Solution **TODO**